

# PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN POSITIVE CHARACTERISTIC

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ABSTRACT. Let  $K$  be a nontrivially normed field and let  $k \geq 0$ . We characterize when precomposition by a continuous linear map is analytic on spaces of continuous alternating  $k$ -linear maps: it is analytic for every triple of  $K$ -Banach spaces if and only if  $k! \neq 0$  in  $K$ . If  $K$  has characteristic  $p > 0$  and  $k \geq p$ , we construct Banach spaces  $E$  and  $F$  for which precomposition on  $\text{Alt}^k(E; F)$  is analytic at no point, although it is of class  $C^n$  for every  $n \in \mathbb{N} \cup \{\infty\}$ . The field  $K$  need not be complete, and  $F$  admits no equivalent ultrametric norm. For ultrametric spherically complete targets, we instead prove analyticity in every degree by constructing a contracting retraction onto alternating maps. The algebraic obstruction is a finite-field theorem: coordinatewise multiplication on sequences has no multilinear lift to the exterior power satisfying the polarized diagonal identity and a uniform bound on support dimension when  $k! = 0$ . We prove this by Ramsey’s theorem and a cancellation argument. A constant-coefficient map on a completed exterior power over a Laurent series field converts boundedness into the required support bound; Ingleton’s Hahn–Banach theorem then permits passage to an arbitrary base field. The main theorem and the factorial criterion are formalized in Lean with Mathlib.

## 1. INTRODUCTION

Let  $K$  be a nontrivially normed field. For normed  $K$ -spaces  $E, E', F$  and a finite set  $\iota$ , write  $\mathcal{L}(E, E')$  for the continuous linear maps with the operator norm and  $\text{Alt}^\iota(E; F)$  for the continuous alternating  $\iota$ -linear maps  $E^\iota \rightarrow F$ . Here *alternating* means that the map vanishes whenever two arguments coincide, including in characteristic 2. We study

$$(1.1) \quad \begin{aligned} Q &= Q_{E, E', F}^K: \mathcal{L}(E, E') \longrightarrow \mathcal{L}(\text{Alt}^\iota(E'; F), \text{Alt}^\iota(E; F)), \\ Q(f)(m) &= m \circ (f, \dots, f). \end{aligned}$$

For multilinear maps without the alternation condition, precomposition is the diagonal of a bounded multilinear map and is therefore analytic. Requiring the values to remain alternating changes the problem. The usual polarization argument divides by  $k!$ , where  $k = |\iota|$ , and does not apply when that scalar vanishes. We show that this obstruction is unavoidable if analyticity is required for all Banach spaces over  $K$ .

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The result, its informal proof and the original write-up were produced with Anthropic’s Claude (in Claude Code), and the informal proofs were audited by independent AI agents. The Main Theorem and its supporting development have since been formalized in Lean with Mathlib using Codex and Autoform. Neither the paper nor the formalization has yet received a human expert review. See §10 for the verification scope and provenance.

**Main results.** We use *Banach space* to mean a complete normed space; its norm need not satisfy the ultrametric inequality. This convention is essential to the result. The base field itself need not be complete.

**Theorem 1** (Main Theorem). *Let  $K$  be a nontrivially normed field of characteristic  $p > 0$ , and let  $k \geq p$ . There are Banach spaces  $E$  and  $F$  over  $K$  such that, for every finite set  $\iota$  with  $|\iota| = k$ , the map*

$$Q: \mathcal{L}(E, E) \rightarrow \mathcal{L}(\text{Alt}^\iota(E; F), \text{Alt}^\iota(E; F)), \quad Q(f)(m) = m \circ (f, \dots, f),$$

*is analytic at no point of  $\mathcal{L}(E, E)$ . Equivalently, there is no bounded  $k$ -linear map  $P$  on  $\mathcal{L}(E, E)$  with  $P(f, \dots, f) = Q(f)$  for all  $f$ . The Banach space  $F$  admits no equivalent ultrametric norm.*

**Corollary 2.** *Let  $K$  be any nontrivially normed field and  $\iota$  any finite set. Then  $Q_{E, E', F}^K$  is analytic for all  $K$ -Banach spaces  $E, E', F$  if and only if  $|\iota|! \neq 0$  in  $K$ . More precisely:*

- (1) *if  $|\iota|! \neq 0$  in  $K$ , then  $Q_{E, E', F}^K$  is analytic, indeed given at every point by a finite power series, for all normed spaces  $E, E', F$ , complete or not;*
- (2) *if  $|\iota|! = 0$  in  $K$ , then there are Banach spaces  $E = E'$  and  $F$ , with  $F$  admitting no equivalent ultrametric norm, such that  $Q_{E, E', F}^K$  is analytic at no point.*

Thus the factorial condition is exact, even when the source and target of the linear maps are required to coincide. The failure is also local in the strongest sense: in the examples, analyticity fails at every base point. By the lift criterion of Proposition 8, it is enough to prove that no bounded multilinear lift exists.

Finite-dimensional source spaces give a second positive case: over a complete field,  $Q$  is analytic in every degree if either  $E$  or  $E'$  is finite-dimensional, with no restriction on  $F$  (Corollary 16). We give explicit determinant formulas for these lifts in §4. Finite-dimensional targets require a different analysis, given in Remark 50.

**Categorical formulation.** Let  $\text{Vec}_K$  be the category of normed  $K$ -spaces, with ordinary, not necessarily ultrametric norms, and bounded linear maps. Write  $\text{Ban}_K$  for its full subcategory of complete spaces, and  $\text{Vec}_K^\circ$  for its full subcategory of ultrametric, spherically complete spaces. For a fixed finite set  $\iota$ , the bifunctor  $\text{Alt}^\iota$  acts on morphisms by

$$(u, v) \mapsto [m \mapsto v \circ m \circ (u, \dots, u)].$$

We call this bifunctor *analytic* if, for every fixed choice of objects  $E, E', F, F'$ , this map

$$\mathcal{L}(E, E') \times \mathcal{L}(F, F') \longrightarrow \mathcal{L}(\text{Alt}^\iota(E'; F), \text{Alt}^\iota(E; F'))$$

is analytic for the operator norms. Corollary 2 and Theorem A give the following categorical statements, with  $k = |\iota|$ .

- (1) The bifunctor

$$\text{Alt}^\iota: \text{Vec}_K^{\text{op}} \times \text{Vec}_K \longrightarrow \text{Vec}_K$$

is analytic if and only if  $k! \neq 0$  in  $K$ . The same criterion holds for its restriction  $\text{Alt}^\iota: \text{Ban}_K^{\text{op}} \times \text{Ban}_K \rightarrow \text{Ban}_K$ . In characteristic  $p > 0$ , the criterion is exactly  $k < p$ .

- (2) If  $K$  is ultrametric, in any characteristic, then

$$\text{Alt}^\iota: \text{Vec}_K^{\text{op}} \times \text{Vec}_K^\circ \longrightarrow \text{Vec}_K^\circ$$

is analytic for every finite  $\iota$ .

Indeed, write  $v_*a = v \circ a$ . The map  $(v, T) \mapsto v_* \circ T$  is bounded bilinear, with  $\|v_* \circ T\| \leq \|v\| \|T\|$ , and the displayed morphism map is  $(u, v) \mapsto v_* \circ Q_{E, E', F}^K(u)$ . Corollary 2 and Theorem A therefore give the positive assertions. Conversely, fixing  $v = \text{id}_F$  recovers  $Q$ , so Corollary 2 gives the necessity in (1), even on Banach spaces. The target categories are correct because  $\text{Alt}^l(E; F)$  is complete whenever  $F$  is complete, and Proposition 4 gives the stronger closure property in (2). Neither statement assumes completeness of  $K$ .

**Motivation and related results.** The regularity of  $Q$  arises in the construction of bundles of differential forms. Applying  $\text{Alt}^l$  to the transition maps of vector bundles requires control of the corresponding precomposition map; compare Lang’s functors of class  $C^p$  [4, Ch. III, §4]. Mathlib treats manifolds and vector bundles over arbitrary nontrivially normed fields, with regularity  $n \in \mathbb{N} \cup \{\infty, \omega\}$ , where  $C^\omega$  denotes analytic regularity. This makes the distinction between smoothness and analyticity in positive characteristic relevant to the general bundle construction.

Over every such field,  $Q$  is of class  $C^n$  for every  $n \in \mathbb{N} \cup \{\infty\}$ , with no completeness or dimension assumptions; this is proved in the draft Mathlib contribution [7]. In characteristic 0, Gouëzel proves analyticity by polarization [1] and uses it in the bundle construction [2]. The same polarization argument works whenever  $k! \neq 0$  in  $K$ ; we prove it in §4. Theorem 1 shows that the regularity range in [7] is sharp over every field of characteristic  $p > 0$  once  $k \geq p$ : in these examples, Mathlib’s `ContDiffAt K ω` fails at every point.

The answer is different if the target norm is required to be ultrametric. If  $K$  is ultrametric and  $F$  is ultrametric and spherically complete, then  $Q$  is analytic for all normed  $E, E'$  (Theorem A). The proof in §2 constructs a contracting retraction from multilinear maps to alternating maps by spherical completeness and extension of bounded linear maps. This positive result is not used in the Main Theorem or Corollary 2. A special case appears in [6, Remark 16]. Over a discretely valued base, Theorem A in fact rules out a Banach counterexample whenever  $F$  has an equivalent ultrametric norm (Proposition 51).

The earlier characteristic-2 construction [6] uses an ultrametric target and gives nonanalyticity at 0 for  $E = E' = \ell^\infty$  and  $k = 2$  over every complete ultrametric field of characteristic 2 that is not spherically complete. Its formal hypothesis is a descending sequence of closed balls with empty intersection, equivalent to failure of spherical completeness (Remark 36). The higher-characteristic cases were left open there. The present construction works over every nontrivially normed field of positive characteristic, including spherically complete fields, by allowing targets with no equivalent ultrametric norm. The remaining questions for ultrametric targets are discussed in §9.

**Overview of the counterexample proof.** The proof separates into an algebraic obstruction, its realization by a normed space, and passage to the prescribed field. The analytic input is a lift criterion valid in every characteristic: if  $Q$  is analytic at any point, the degree- $k$  term of its power series has diagonal  $Q$ . One-variable uniqueness identifies this diagonal without dividing by  $k!$  (Proposition 8).

*The finite-field obstruction.* Let  $\kappa$  be finite and put  $E_0 = \kappa^\mathbb{N}$ . Theorem 23 rules out a multilinear lift of coordinatewise multiplication satisfying the polarized diagonal identity (Pol1), with values in  $\Lambda_\kappa^k E_0$  when  $k! = 0$  and every value of the lift lies in an exterior power of a subspace of uniformly bounded dimension. Ramsey’s theorem first makes the coefficients depend only on the order and equality pattern of their indices. Signed sums over clusters of four indices cancel the terms not tied to the output. A staircase of clusters converts the support bound into invariance under adjacent transpositions; the polarized diagonal identity then forces  $k! \chi = 1$  for a common cluster value  $\chi$ , a contradiction.

*The Laurent series construction.* Over  $K_1 = \kappa((X))$ , take  $E_1 = \ell^\infty(\mathbb{N}, K_1)$  and let  $B$  be the completion of  $\Lambda_{K_1}^k E_1$  under its projective norm, defined using ordinary sums rather than maxima. The essential estimate is that the constant-coefficient map

$$\eta: B \longrightarrow \Lambda_{\kappa}^k E_0$$

is well defined and has uniformly bounded support dimension on bounded sets (Proposition 31). Evaluating a hypothetical lift on multiplication operators and on the universal wedge map, then applying  $\eta$ , produces exactly the finite-field lift excluded above. The coefficient map is  $\kappa$ -linear; no  $K_1$ -linearity is asserted or needed.

*Passage to the prescribed field.* Choose  $t \in K$  with  $0 < |t| < 1$ . The closure of  $\mathbb{F}_p(t)$  in the completion  $\widehat{K}$  is a spherically complete Laurent series field  $K_1 \cong \mathbb{F}_p((X))$ , with  $|X| = |t|$ . Ingleton's Hahn–Banach theorem [3] gives a contracting  $K_1$ -linear projection  $\widehat{K} \rightarrow K_1$ . It makes the completed projective tensor products

$$E = E_1 \widehat{\otimes}_{\pi, K_1} \widehat{K}, \quad F = B \widehat{\otimes}_{\pi, K_1} \widehat{K}$$

contain  $E_1$  and  $B$  isometrically, and allows a lift over  $\widehat{K}$  to descend to one over  $K_1$ . Density handles an incomplete  $K$ . These are the explicit witnesses in Theorem 1. Finally, sums of unit wedges on disjoint blocks have norms growing linearly with the number of blocks. This growth persists in  $F$  and rules out every equivalent ultrametric norm (Proposition 48).

**Formal verification and organization.** The Main Theorem, both directions of Corollary 2, and their supporting arguments are formalized in Lean 4 with Mathlib. The development also includes Theorem A, the discrete-target result, and the sorted-basis lift of §9. The additional finite-dimensional arguments and their formalization status are identified in §10. §10 records the principal declarations, verification scope, and provenance; the quoted smoothness result [7], whose argument we include, and earlier characteristic-2 construction [6] have separate provenance.

§2 proves Theorem A independently, including the extension and spherical-completeness arguments. §3 gives the analytic and exterior-algebra preliminaries, and §4 proves the positive cases. §5 proves the finite-field obstruction, and §6 realizes it over a Laurent series field. §7 develops scalar extension and descent; §8 assembles the main results. §9 treats ultrametric targets and open problems.

## 2. SPHERICALLY COMPLETE TARGETS: PROOF OF THEOREM A

We first prove a positive result that explains the role of the target norm. Spherical completeness allows us to project bounded multilinear maps onto alternating maps without increasing the norm. Applying this projection to the ordinary multilinear precomposition formula produces an analytic lift in every characteristic and degree. The argument is independent of the counterexample construction. A special case is stated in [6, Remark 16]; the proof here follows the extension and retraction arguments in the Lean development.

**2.1. Statement and strategy.** A norm is *ultrametric* if  $\|x + y\| \leq \max(\|x\|, \|y\|)$ . A metric space is *spherically complete* if every nonempty family of closed balls that meet pairwise has a common point. We write  $\overline{B}(a, r) = \{x : \|x - a\| \leq r\}$ , with  $r \geq 0$ . In an ultrametric space, two such balls meet if and only if the distance between their centres is at most the larger radius; in that case the ball of smaller radius is contained in the other.

As in the introduction,  $\mathcal{L}(X, Y)$  denotes bounded linear maps. Write  $\text{Mult}^k(X; Y)$  for the bounded  $k$ -linear maps and  $\text{Alt}^k(X; Y)$  for their alternating subspace, with the operator norms

determined by

$$\|m(x_1, \dots, x_k)\| \leq \|m\| \prod_{i=1}^k \|x_i\|.$$

When the target norm is ultrametric, both operator spaces are ultrametric, regardless of the norm on  $X$ . For a finite set  $\iota$  of size  $k$ , choosing a bijection with  $\{1, \dots, k\}$  identifies the corresponding spaces isometrically and preserves precomposition. We therefore use the degree notation  $\text{Alt}^k$  in the proof.

**Theorem A.** Let  $K$  be an ultrametric nontrivially normed field, let  $E, E'$  be normed  $K$ -spaces, and let  $F$  be a normed  $K$ -space whose norm is ultrametric and which is spherically complete as a metric space. Then  $Q_{E, E', F}^K$  is  $C^n$  for every  $n \in \mathbb{N} \cup \{\infty, \omega\}$ ; in particular it is analytic at every point.

No completeness assumption on  $K, E$  or  $E'$  is needed, and the norms on  $E, E'$  may be non-ultrametric. For  $k \geq 1$ , put  $A = \text{Alt}^k(E; F)$  and  $M = \text{Mult}^k(E; F)$ . The desired projection is a bounded linear map  $\rho: M \rightarrow A$  of norm at most one that fixes every alternating map. Once it is available, the formula

$$P(f_1, \dots, f_k)(m) = \rho(m \circ (f_1, \dots, f_k))$$

is bounded and multilinear in the operators  $f_i$ , and has diagonal  $Q$ . This is the required analytic lift: expansion around any operator gives a finite power series. To construct  $\rho$ , we first prove an extension lemma that preserves a family of bounds, then use it to show that  $A$  is spherically complete.

**2.2. Extension with simultaneous bounds.** The following form of the ultrametric extension argument keeps a prospective extension in several operator balls at once. We allow a seminorm on the domain, since the free space used in the next step may have nonzero vectors of seminorm zero.

**Lemma 3** (compatible extension). *Let  $U$  be an ultrametric seminormed  $K$ -space and let  $G$  be an ultrametric, spherically complete normed  $K$ -space. Suppose a nonempty family  $(T_\alpha, r_\alpha)$ , with  $T_\alpha \in \mathcal{L}(U, G)$  and  $r_\alpha \geq 0$ , satisfies*

$$\|T_\alpha - T_\beta\| \leq \max(r_\alpha, r_\beta) \quad \text{for all } \alpha, \beta.$$

*Let  $D \subseteq U$  be a linear subspace and let  $T_0: D \rightarrow G$  be linear with*

$$\|T_0 d - T_\alpha d\| \leq r_\alpha \|d\| \quad (d \in D, \text{ all } \alpha).$$

*Then  $T_0$  extends to a bounded linear map  $T: U \rightarrow G$  such that  $\|T - T_\alpha\| \leq r_\alpha$  for every  $\alpha$ . In particular, every contraction from a subspace of  $U$  to  $G$  extends to a contraction on  $U$ .*

*Proof.* It suffices to extend across one vector  $x \notin D$  while preserving every bound. The required value  $Tx$  must belong to the balls

$$(2.1) \quad \overline{B}(T_\alpha(d+x) - T_0 d, r_\alpha \|d+x\|) \quad (\alpha, d \in D).$$

We check that these balls meet pairwise. Write  $c_{\alpha,d} = T_\alpha(d+x) - T_0 d$ , and suppose  $r_\alpha \leq r_\beta$ . Then

$$c_{\alpha,d} - c_{\beta,e} = T_\alpha(d-e) - T_0(d-e) + (T_\alpha - T_\beta)(e+x).$$

Since  $d-e \in D$  and  $\|d-e\| \leq \max(\|d+x\|, \|e+x\|)$ , the hypotheses give

$$\begin{aligned} \|c_{\alpha,d} - c_{\beta,e}\| &\leq \max(r_\alpha \|d-e\|, r_\beta \|e+x\|) \\ &\leq \max(r_\alpha \|d+x\|, r_\beta \|e+x\|). \end{aligned}$$

This is exactly the condition for the two balls to meet. The other ordering of the radii is symmetric.

By spherical completeness, choose  $c$  in all the balls (2.1). Define

$$T(d + \lambda x) = T_0 d + \lambda c.$$

The decomposition is unique because  $x \notin D$ . For  $\lambda \neq 0$ , apply the ball condition with  $\lambda^{-1}d$  in place of  $d$  and multiply by  $|\lambda|$ ; this gives  $\|T(d + \lambda x) - T_\alpha(d + \lambda x)\| \leq r_\alpha \|d + \lambda x\|$ . For  $\lambda = 0$  the original bound applies.

Partially order the extensions satisfying all these inequalities by inclusion of their domains. Unions of chains remain linear and preserve the bounds, so Zorn's lemma gives a maximal extension. The one-vector argument forces its domain to be all of  $U$ . Fixing any one index  $\alpha$  gives  $\|Tu\| \leq \max(\|T_\alpha\|, r_\alpha)\|u\|$ , so the extension is bounded. For the last assertion, take the single pair  $(T_\alpha, r_\alpha) = (0, 1)$ .  $\square$

**2.3. Spherical completeness of alternating-map spaces.** Ordinary completeness alone would not let us apply the extension lemma with target  $A$ . We prove the required stronger property by presenting alternating maps as linear maps on a free space subject to linear relations. The ultrametric seminorm on that free space is chosen so that this presentation preserves the operator norm, even when  $E$  itself is not ultrametric.

**Proposition 4.** *Under the hypotheses of Theorem A,  $\text{Alt}^k(E; F)$  is an ultrametric, spherically complete normed space for every  $k \geq 0$ .*

*Proof.* For  $k = 0$ , the space identifies isometrically with  $F$ . Suppose  $k \geq 1$ . Let  $U$  be the free  $K$ -vector space on the set  $E^k$ , with generators  $[x]$  and seminorm

$$\left\| \sum_x c_x [x] \right\|_U = \max_x |c_x| \prod_{i=1}^k \|x_i\|.$$

The maximum is over the finite support, with empty maximum zero. This seminorm is ultrametric because the absolute value on  $K$  is ultrametric. Let  $R \subseteq U$  be the linear span of the relations expressing additivity and scalar homogeneity in each slot, together with  $[x]$  whenever two coordinates of  $x$  coincide.

An alternating map  $a \in A$  defines a linear map vanishing on  $R$  by

$$\widehat{a} \left( \sum_x c_x [x] \right) = \sum_x c_x a(x).$$

The ultrametric inequality in  $F$  gives

$$\left\| \widehat{a} \left( \sum_x c_x [x] \right) \right\| \leq \max_x |c_x| \|a(x)\| \leq \|a\| \left\| \sum_x c_x [x] \right\|_U.$$

Conversely, a bounded linear map  $T: U \rightarrow F$  vanishing on  $R$  defines an alternating map  $a(x) = T([x])$  with  $\|a(x)\| \leq \|T\| \prod_i \|x_i\|$ . Thus these two constructions are inverse linear isometries. In particular, the equal-input relations impose strong alternation in every characteristic.

Take a nonempty family of pairwise-meeting closed balls in  $A$ , with centres  $a_\alpha$  and radii  $r_\alpha$ . Their corresponding linear maps satisfy  $\|\widehat{a}_\alpha - \widehat{a}_\beta\| \leq \max(r_\alpha, r_\beta)$  and vanish on  $R$ . Apply Lemma 3 with  $D = R$ ,  $T_0 = 0$  and  $G = F$ . It produces a bounded linear map  $T: U \rightarrow F$  that vanishes on  $R$  and satisfies  $\|T - \widehat{a}_\alpha\| \leq r_\alpha$  for every  $\alpha$ . The associated alternating map belongs to every original ball. This proves spherical completeness; no closure assumption on  $R$  is needed.  $\square$

**2.4. The retraction and the analytic lift.** We now apply the extension lemma a second time, using the alternating-map space itself as the target. This produces the projection needed to turn ordinary precomposition into a polynomial with alternating values.

*Proof of Theorem A.* In degree zero, precomposition is constant. Assume  $k \geq 1$  and retain  $A = \text{Alt}^k(E; F)$  and  $M = \text{Mult}^k(E; F)$ . Their operator norms are ultrametric, and  $A$  is spherically complete by Proposition 4. The inclusion  $j: A \rightarrow M$  is a linear isometry. Its inverse on  $j(A)$  is therefore a contraction. Lemma 3 extends this inverse to a contraction  $\rho: M \rightarrow A$  with  $\rho j = \text{id}_A$ .

For  $f_1, \dots, f_k \in \mathcal{L}(E, E')$  and  $m \in \text{Alt}^k(E'; F)$ , define

$$P(f_1, \dots, f_k)(m) = \rho(m \circ (f_1, \dots, f_k)).$$

This is linear in  $m$  and multilinear in the  $f_i$ . The estimate

$$\|P(f_1, \dots, f_k)(m)\| \leq \|m\| \prod_{i=1}^k \|f_i\|$$

shows that  $P$  is a bounded  $k$ -linear map into  $\mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$ , with  $\|P\| \leq 1$ . On the diagonal,  $m \circ (f, \dots, f)$  is already alternating, so  $P(f, \dots, f) = Q(f)$ .

For completeness, expand at an arbitrary  $f_0 \in \mathcal{L}(E, E')$ . For  $0 \leq r \leq k$ , define a bounded  $r$ -linear map  $q_r$  by summing over the  $r$ -element subsets  $S$  of  $\{1, \dots, k\}$ : insert its  $r$  arguments in the slots of  $S$ , in increasing order, and put  $f_0$  in all remaining slots of  $P$ . Multilinearity gives the finite identity

$$Q(f_0 + h) = \sum_{r=0}^k q_r(h, \dots, h), \quad \|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}.$$

The binomial coefficient in the bound is an ordinary nonnegative integer counting the summands. With  $q_r = 0$  for  $r > k$ , this is a power series of infinite radius at every  $f_0$ . Consequently  $Q$  is continuously polynomial, hence  $C^\omega$  and  $C^m$  for all  $n \in \mathbb{N} \cup \{\infty\}$ , as asserted.  $\square$

The retraction supplies the alternating values directly, with no division by  $k!$ . The source spaces need no ultrametric hypothesis because the extension argument takes place in the free presentation and in operator spaces with ultrametric targets. The following sections develop the other positive cases and then construct counterexamples whose targets admit no equivalent ultrametric norm.

### 3. ANALYTIC AND ALGEBRAIC PRELIMINARIES

The lift criterion reduces the analytic question to a bounded multilinear identity. We then introduce support dimension and the determinant arrays that detect it; these provide the rank estimates used in the finite-field argument.

**3.1. Conventions.** We write  $\mathbb{N} = \{0, 1, 2, \dots\}$ . Throughout,  $K$  denotes the base field and  $k$  the degree. A *normed field*  $K$  carries a multiplicative absolute value  $|\cdot|$  satisfying the triangle inequality. It is *nontrivially normed* if some  $c \in K$  has  $0 < |c| < 1$ , and *ultrametric* if  $|x + y| \leq \max(|x|, |y|)$ . A *normed  $K$ -space*  $X$  satisfies  $\|\lambda x\| = |\lambda| \|x\|$ ; it is *Banach* if it is complete. The norm of a space need not be ultrametric. In non-archimedean functional analysis, the term “Banach” often includes the ultrametric condition; Theorem A and §9 address ultrametric targets.

We write  $\mathcal{L}(X, Y)$  for the bounded linear maps,  $\text{Mult}^k(X; Y)$  for the bounded  $k$ -linear maps  $X^k \rightarrow Y$ , and  $\text{Alt}^k(X; Y) \subseteq \text{Mult}^k(X; Y)$  for the closed subspace of alternating maps, all

with the operator norm. A subscript  $K$  records the field when needed. An alternating map  $m$  is antisymmetric: expanding  $0 = m(\dots, y + z, \dots, y + z, \dots)$  gives  $m(\dots, y, \dots, z, \dots) = -m(\dots, z, \dots, y, \dots)$ , and hence  $m(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \text{sgn}(\sigma) m(x_1, \dots, x_k)$  for  $\sigma \in S_k$ . In characteristic 2 the converse fails.

For a finite set  $\iota$  with  $|\iota| = k$ , a bijection  $\iota \cong \{1, \dots, k\}$  identifies  $\text{Alt}^t(E; F)$  isometrically with  $\text{Alt}^k(E; F)$  and the map (1.1) with the corresponding map for  $\{1, \dots, k\}$ ; nothing below depends on the choice. So we work with

$$Q = Q_{E, E', F}^K: \mathcal{L}(E, E') \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)),$$

where  $Q(f)(m)(x_1, \dots, x_k) = m(fx_1, \dots, fx_k)$ .

**Definition 5.** A *bounded  $k$ -linear lift* of  $Q$  is a bounded  $k$ -linear map  $P: \mathcal{L}(E, E')^k \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$  with  $P(f, \dots, f) = Q(f)$  for all  $f \in \mathcal{L}(E, E')$ .

A map  $g: X \rightarrow Z$  between normed spaces is *analytic at  $x_0$*  if there are bounded  $n$ -linear maps  $p_n: X^n \rightarrow Z$  ( $n \geq 0$ ) and a radius  $R > 0$  with  $\sup_n \|p_n\| R^n < \infty$  such that  $g(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$ , a convergent series, whenever  $\|h\| < R$ . Up to shrinking  $R$  this is Mathlib's `AnalyticAt` [5], and in Mathlib `ContDiffAt`  $K \omega$  at a point implies `AnalyticAt` there (`ContDiffAt.analyticAt`).

**Lemma 6.** A normed field of characteristic  $p > 0$  is ultrametric.

*Proof.* For  $n \in \mathbb{N}$  the element  $n \cdot 1$  lies in the prime field  $\mathbb{F}_p$ , so  $(n \cdot 1)^p = n \cdot 1$  and  $|n \cdot 1| \in \{0, 1\}$ . Hence  $|(x + y)^N| \leq \sum_{i=0}^N \binom{N}{i} |x|^i |y|^{N-i} \leq (N + 1) \max(|x|, |y|)^N$  for every  $N$ . Taking  $N$ -th roots and letting  $N \rightarrow \infty$  gives  $|x + y| \leq \max(|x|, |y|)$ .  $\square$

**3.2. The lift criterion.** A power series for  $Q$  at any base point determines a lift through its degree- $k$  term. The only uniqueness statement needed is one-dimensional, so the argument does not use a polarization formula or division by  $k!$ .

**Lemma 7** (one-variable uniqueness). *Let  $K$  be nontrivially normed and  $Z$  a normed  $K$ -space, not necessarily complete. Let  $z_0, z_1, \dots \in Z$ ,  $\rho > 0$  and  $M \geq 0$  with  $\|z_n\| \rho^n \leq M$  for all  $n$ . If  $\sum_n t^n z_n$  converges to 0 for every  $t \in K$  with  $|t| < \rho$ , then  $z_n = 0$  for all  $n$ .*

*Proof.* Otherwise let  $N$  be least with  $z_N \neq 0$ . For  $0 < |t| < \rho/2$  the series  $\sum_{n \geq N} t^{n-N} z_n$  converges, being  $t^{-N}$  times the convergent tail  $\sum_{n \geq N} t^n z_n$ , and  $t^N (z_N + \sum_{n \geq N} t^{n-N} z_n) = 0$ . Hence

$$\|z_N\| = \left\| \sum_{n \geq N} t^{n-N} z_n \right\| \leq \sum_{n \geq N} M \rho^{-n} |t|^{n-N} = M \rho^{-N} \frac{|t|/\rho}{1 - |t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since  $K$  is nontrivially normed, there are  $t \neq 0$  with  $|t|$  arbitrarily small, so  $\|z_N\| = 0$ , a contradiction.  $\square$

**Proposition 8** (lift criterion). *Let  $K$  be any nontrivially normed field,  $E, E', F$  normed  $K$ -spaces and  $k \geq 1$ .*

- (1) *If  $Q$  has a bounded  $k$ -linear lift  $P$ , then at every  $f_0 \in \mathcal{L}(E, E')$  the map  $Q$  is given by a finite power series of infinite radius; in particular  $Q$  is analytic at every point.*
- (2) *If  $Q$  is analytic at some point  $f_0$ , with power series  $(p_n)_n$  there, then  $p_k$  is a bounded  $k$ -linear lift of  $Q$ .*
- (3) *Hence the following are equivalent:  $Q$  has a bounded  $k$ -linear lift;  $Q$  is analytic at every point;  $Q$  is analytic at some point. If they fail, then Mathlib's `ContDiffAt`  $K \omega$   $Q$   $f_0$  fails at every  $f_0$ .*

*Proof.* (1) For  $y \in \mathcal{L}(E, E')$ , multilinearity gives

$$Q(f_0 + y) = P(f_0 + y, \dots, f_0 + y) = \sum_{r=0}^k q_r(y, \dots, y),$$

where  $q_r(y_1, \dots, y_r)$  is the sum, over the  $r$ -element subsets  $S \subseteq \{1, \dots, k\}$ , of  $P$  evaluated with  $y_1, \dots, y_r$  in the slots of  $S$  in increasing order and  $f_0$  in the other slots. Each  $q_r$  is bounded  $r$ -linear, with  $\|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}$ , so  $(q_r)_{r \leq k}$  is a power series of infinite radius for  $Q$  at  $f_0$ .

(2) We compare the given power series along each line with the polynomial obtained by expanding precomposition. The expansion is taken in the larger space of all multilinear maps, where each coefficient is visibly defined. Let  $\text{incl}: \text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$  be the isometric inclusion, and put

$$Z := \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)), \quad \mathcal{I}(T) := \text{incl} \circ T.$$

Then  $\mathcal{I}$  is a linear isometry into  $Z$ , hence injective. Fix  $h \in \mathcal{L}(E, E')$ ,  $h \neq 0$ . For  $m \in \text{Alt}^k(E'; F)$ ,  $x \in E^k$  and  $t \in K$ , multilinearity of  $m$  gives the *ambient polynomial identity*

$$(3.1) \quad \mathcal{I}(Q(f_0 + th)) = \sum_{r=0}^k t^r A_r(h), \quad A_r(h)(m)(x) := \sum_{|S|=r} m(u_1^S, \dots, u_k^S),$$

where  $u_i^S = hx_i$  for  $i \in S$  and  $u_i^S = f_0x_i$  for  $i \notin S$ . Here  $A_r(h) \in Z$  is bounded, with  $\|A_r(h)\| \leq \binom{k}{r} \|h\|^r \|f_0\|^{k-r}$ , and  $A_k(h) = \mathcal{I}(Q(h))$ . (For  $r < k$  we do not need to know whether  $A_r(h)(m)$  is alternating; this is why (3.1) is written in  $Z$ .) Let  $R$  and  $M' := \sup_n \|p_n\| R^n$  be as in the definition of analyticity, and put  $\rho := R/\|h\|$ . For  $|t| < \rho$  we have  $\|th\| < R$ , so  $Q(f_0 + th) = \sum_n t^n p_n(h, \dots, h)$ , and applying  $\mathcal{I}$  and (3.1) gives  $\sum_n t^n z_n = 0$  in  $Z$ , with  $z_n := \mathcal{I}(p_n(h, \dots, h)) - A_n(h)$  and  $A_n(h) := 0$  for  $n > k$ . Since  $\|z_n\| \rho^n \leq M' + \max_{r \leq k} \|A_r(h)\| \rho^r$ , Lemma 7 gives  $z_k = 0$ , that is,  $\mathcal{I}(p_k(h, \dots, h)) = \mathcal{I}(Q(h))$ , hence  $p_k(h, \dots, h) = Q(h)$ . For  $h = 0$  both sides vanish since  $k \geq 1$ .

(3) follows from (1) and (2), and from `ContDiffAt.analyticAt` for the last clause.  $\square$

*Remark 9.* The degree- $k$  term of *every* power series representing  $Q$  is a lift, regardless of the base point. This does not assert uniqueness of the multilinear lift: in positive characteristic, different multilinear maps can have the same diagonal.

**3.3. Exterior powers, supports and determinant arrays.** Let  $L$  be a field,  $V$  an  $L$ -vector space and  $k \geq 1$ . We write  $\Lambda^k V = \Lambda_L^k V$  for the algebraic exterior power, in which  $x_1 \wedge \dots \wedge x_k = 0$  whenever two  $x_i$  coincide. We use the standard facts: for a subspace  $W \subseteq V$  the map  $\Lambda^k W \rightarrow \Lambda^k V$  is injective, and we regard  $\Lambda^k W$  as a subspace of  $\Lambda^k V$ ; if  $u_1, \dots, u_d$  is a basis of a finite-dimensional space  $U$ , the wedges  $u_I := u_{i_1} \wedge \dots \wedge u_{i_k}$ ,  $I = \{i_1 < \dots < i_k\}$ , form a basis of  $\Lambda^k U$ ; and every  $\omega \in \Lambda^k V$  lies in  $\Lambda^k U$  for some finite-dimensional  $U \subseteq V$ .

**Definition 10.** The *support dimension* of  $\omega \in \Lambda^k V$  is

$$\text{sdim}(\omega) := \min\{\dim W : W \subseteq V \text{ finite-dimensional, } \omega \in \Lambda^k W\}.$$

The definition gives  $\text{sdim}(0) = 0$ ,  $\text{sdim}(c\omega) = \text{sdim}(\omega)$  for  $c \neq 0$ ,  $\text{sdim}(\omega + \omega') \leq \text{sdim}(\omega) + \text{sdim}(\omega')$  (use  $W + W'$ ), and  $\text{sdim}(\omega) \leq \dim \text{span } S$  if  $\omega$  is a sum of wedges of vectors from a finite set  $S$ . If  $V \subseteq V'$ , then  $\text{sdim}$  computed in  $V'$  is at most  $\text{sdim}$  computed in  $V$ .

For spaces of functions, exterior vectors can be detected by determinants of coordinate evaluations. These arrays also turn bounds on support dimension into ordinary matrix-rank

bounds. Let  $S$  be a set and  $V \subseteq L^S$  a space of functions on  $S$ . The *determinant array* is the linear map

$$\Omega: \Lambda^k V \rightarrow L^{S^k}, \quad \Omega_{y_1 \wedge \dots \wedge y_k}(c_1, \dots, c_k) := \det(y_b(c_a))_{a,b=1}^k.$$

It is well defined because the right side is multilinear and alternating in  $(y_1, \dots, y_k)$  (these are the columns). For each  $\omega$ , the function  $\Omega_\omega$  is alternating in  $(c_1, \dots, c_k)$  (the rows); in particular  $\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)}) = \text{sgn}(\sigma) \Omega_\omega(c_1, \dots, c_k)$ .

**Lemma 11.** *The determinant array  $\Omega: \Lambda^k V \rightarrow L^{S^k}$  is injective.*

*Proof.* Let  $\Omega_\omega = 0$  and choose a finite-dimensional  $U \subseteq V$  with  $\omega \in \Lambda^k U$ . The evaluations  $\text{ev}_s|_U$ ,  $s \in S$ , separate the points of  $U$  (a function in  $U$  vanishing at every  $s$  is 0), so they span the dual space  $U^*$ . Choose  $s_1, \dots, s_d \in S$  such that the  $\text{ev}_{s_i}|_U$  form a basis of  $U^*$ , and let  $u_1, \dots, u_d$  be the dual basis of  $U$ , so that  $u_j(s_i) = \delta_{ij}$ . For increasing  $k$ -subsets  $I, J$  of  $\{1, \dots, d\}$  we get  $\Omega_{u_I}(s_{j_1}, \dots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$ . Writing  $\omega = \sum_I c_I u_I$  therefore gives  $0 = \Omega_\omega(s_{j_1}, \dots, s_{j_k}) = c_J$  for every  $J$ , so  $\omega = 0$ .  $\square$

**Lemma 12** (support bounds flattening rank). *Let  $V \subseteq L^S$ ,  $\omega \in \Lambda^k V$  and  $\alpha \in \{1, \dots, k\}$ . Let  $R \subseteq S$  and  $R' \subseteq S^{\{1, \dots, k\} \setminus \{\alpha\}}$  be finite. The  $R \times R'$  matrix  $M$  with entries  $M(c_\alpha, c') := \Omega_\omega(c)$ , where  $c$  has  $c_\alpha$  in slot  $\alpha$  and the entries of  $c'$  in the other slots, has rank at most  $\text{sdim}(\omega)$ .*

*Proof.* Write  $\omega = \sum_j y_{j,1} \wedge \dots \wedge y_{j,k}$  with all  $y_{j,b}$  in a subspace  $W$  of dimension  $\text{sdim}(\omega)$ . Expanding each determinant along row  $\alpha$ ,  $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot \mu_{j,b}(c')$ , where the minor  $\mu_{j,b}(c')$  does not involve  $c_\alpha$ . So for fixed  $c'$  the column  $c_\alpha \mapsto M(c_\alpha, c')$  is a linear combination of the restrictions  $y_{j,b}|_R$ . All columns lie in the restriction of  $W$  to  $R$ , a space of dimension at most  $\text{sdim}(\omega)$ .  $\square$

A second way to obtain lower bounds on support is to contract all but one exterior factor. The resulting vectors must lie in every subspace supporting the original exterior vector.

**Lemma 13** (contraction). *Let  $k \geq 2$  and  $\varphi = (\varphi_1, \dots, \varphi_{k-1}) \in (V^*)^{k-1}$ . There is a linear map  $c_\varphi: \Lambda^k V \rightarrow V$  with*

$$c_\varphi(x_1 \wedge \dots \wedge x_k) = \sum_{i=1}^k (-1)^{k+i} \det(\varphi_a(x_b))_{1 \leq a \leq k-1, b \neq i} x_i,$$

*and  $c_\varphi(\Lambda^k W) \subseteq W$  for every subspace  $W$ . Consequently  $\text{sdim}(\omega) \geq \dim \text{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$ .*

*Proof.* The right side is multilinear in  $(x_b)$ . For  $\psi \in V^*$ , applying  $\psi$  to it gives, by Laplace expansion along the last row, the determinant of the  $k \times k$  matrix with entries  $\varphi_a(x_b)$  in the rows  $a < k$  and  $\psi(x_b)$  in the last row; it vanishes when two columns coincide. Since  $V^*$  separates the points of  $V$ , the right side vanishes whenever  $x_a = x_b$  for some  $a \neq b$ , so it factors through  $\Lambda^k V$ . Each term is a multiple of some  $x_i$ , which gives  $c_\varphi(\Lambda^k W) \subseteq W$ . If  $\omega \in \Lambda^k W$  with  $\dim W = \text{sdim}(\omega)$ , all  $c_\varphi(\omega)$  lie in  $W$ .  $\square$

#### 4. POSITIVE RESULTS

Alongside the spherical-completeness argument of §2, two further constructions give analytic lifts: dividing by  $k!$ , when this scalar is nonzero, or choosing coordinates under suitable finite-dimensional hypotheses. We also prove smoothness in every characteristic. It descends from the space of all multilinear maps through a closed isometric inclusion and therefore requires neither hypothesis.

#### 4.1. Invertible factorial: in particular, $p > k$ .

**Proposition 14.** *Let  $K$  be a nontrivially normed field and let  $E, E', F$  be normed  $K$ -spaces. If  $k! \neq 0$  in  $K$ , then  $Q$  is given at every point by a finite power series of infinite radius. In particular, it is analytic whenever  $\text{char } K = 0$ , or whenever  $\text{char } K = p > k$ . No completeness or dimension hypothesis is needed.*

*Proof.* For  $k = 0$ , precomposition is constant. Suppose  $k \geq 1$  and define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \frac{1}{k!} \sum_{\sigma \in S_k} m(f_{\sigma(1)}x_1, \dots, f_{\sigma(k)}x_k).$$

This is multilinear in the  $f_i$  and in the  $x_j$ , linear in  $m$ , and satisfies

$$\|P(f_1, \dots, f_k)(m)(x_1, \dots, x_k)\| \leq |(k!)^{-1}| k! \|m\| \prod_i \|f_i\| \prod_j \|x_j\|,$$

where the second  $k!$  on the right is a real number counting the terms. To check alternation, suppose  $x_a = x_b$  and let  $\tau$  exchange  $a$  and  $b$ . The summands indexed by  $\sigma$  and  $\sigma \circ \tau$  cancel by antisymmetry of  $m$ . This pairing is a fixed-point-free involution, so it proves vanishing also in characteristic 2. Thus  $P$  takes values in the required space of alternating maps and is a bounded  $k$ -linear lift: on the diagonal, all  $k!$  summands equal  $Q(f)(m)(x)$ . Proposition 8 gives the claimed power series.

In a field of characteristic  $p > 0$ , the scalar  $k!$  is nonzero precisely when  $k < p$ . This gives the stated positive-characteristic case.  $\square$

**4.2. Finite-dimensional domain or codomain.** The finite-dimensional argument uses continuous coordinate functionals. We state this hypothesis explicitly so that the result also applies over incomplete fields whenever the chosen finite-dimensional space has its usual coordinate topology.

**Proposition 15** (finite coordinate bases). *Let  $K$  be a nontrivially normed field and let  $E, E', F$  be normed  $K$ -spaces. If either  $E$  or  $E'$  has a finite algebraic basis whose coordinate functionals are continuous, then  $Q_{E, E', F}^K$  has a bounded  $k$ -linear lift in every degree  $k \geq 1$ . In particular, it is continuously polynomial, and hence analytic, at every point. No completeness or ultrametric hypothesis on  $F$  is required.*

*Proof.* Let  $H$  denote the space with a finite coordinate basis, write that basis as  $e_1, \dots, e_n$ , and let  $\varepsilon_1, \dots, \varepsilon_n \in \mathcal{L}_K(H, K)$  be its coordinate functionals. For an increasing  $k$ -tuple  $s = (s_1 < \dots < s_k)$  in  $\{1, \dots, n\}$ , put

$$c_s := \prod_{a=1}^k \|\varepsilon_{s_a}\| \|e_{s_a}\|, \quad C_H := k! \sum_{s_1 < \dots < s_k} c_s.$$

Here the factor  $k!$  in  $C_H$  is the positive real integer counting the terms in a determinant, not its image in  $K$ . In particular,  $C_H \geq 0$  even when  $k! = 0$  in  $K$ .

Suppose first that  $H = E$ . For  $f_1, \dots, f_k \in \mathcal{L}_K(E, E')$ ,  $m \in \text{Alt}_K^k(E'; F)$  and  $x \in E^k$ , define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(x_b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

The determinant makes this expression alternating and multilinear in  $x$ . It is linear in  $m$  and multilinear in  $f_1, \dots, f_k$ . Expanding the determinant over permutations gives

$$\left| \det(\varepsilon_{s_a}(x_b))_{a,b} \right| \leq k! \prod_a \|\varepsilon_{s_a}\| \prod_b \|x_b\|.$$

Together with the operator bound for  $m$ , the ordinary triangle inequality therefore yields

$$\|P(f_1, \dots, f_k)(m)(x)\| \leq C_E \|m\| \prod_a \|f_a\| \prod_b \|x_b\|.$$

Thus the displayed formula defines a bounded  $k$ -linear map into  $\mathcal{L}_K(\text{Alt}_K^k(E'; F), \text{Alt}_K^k(E; F))$ , with  $\|P\| \leq C_E$ . On the diagonal, the finite exterior-basis expansion of  $x_1 \wedge \dots \wedge x_k$  gives

$$m(fx_1, \dots, fx_k) = \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(x_b))_{a,b} m(fe_{s_1}, \dots, fe_{s_k}),$$

so  $P(f, \dots, f) = Q(f)$ .

Now suppose that  $H = E'$ . Define instead

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(f_a x_b))_{a,b} m(e_{s_1}, \dots, e_{s_k}).$$

Each row of the determinant is linear in its corresponding operator  $f_a$ , and its columns make the result alternating and multilinear in  $x$ . The expression is again linear in  $m$ . The determinant estimate now reads

$$\left| \det(\varepsilon_{s_a}(f_a x_b))_{a,b} \right| \leq k! \prod_a (\|\varepsilon_{s_a}\| \|f_a\|) \prod_b \|x_b\|.$$

It follows that the same operator bound holds with  $C_{E'}$  in place of  $C_E$ . Expanding  $fx_1 \wedge \dots \wedge fx_k$  in the exterior basis of  $E'$  proves the diagonal identity.

Both formulas are finite sums. Strong alternation follows from equal columns in a determinant, so the proof works in every characteristic and involves no division by  $k!$ . Proposition 8 gives continuous polynomiality and analyticity. If  $k > n$ , the sums are empty and the relevant alternating-map space is zero; the same argument gives the zero lift. Degree zero, excluded from the lift criterion's statement, is immediate because  $Q$  is constant.  $\square$

**Corollary 16.** *Let  $K$  be a complete nontrivially normed field, and let  $E, E', F$  be normed  $K$ -spaces. If either  $E$  or  $E'$  is finite-dimensional over  $K$ , then  $Q_{E, E', F}^K$  is analytic at every point in every degree. No completeness assumption on  $E, E'$  or  $F$  is needed in the statement.*

*Proof.* We recall why a finite-dimensional normed space  $H$  over complete  $K$  has continuous coordinates. Induct on its dimension, proving at the same time that  $H$  is complete. The zero-dimensional case is immediate. For a basis  $e_1, \dots, e_n$ , the subspace  $W := \text{span}(e_1, \dots, e_{n-1})$  is complete by induction, hence closed in  $H$ . Since  $e_n \notin W$ , the distance  $\delta := \inf_{w \in W} \|e_n - w\|$  is positive. If  $x = w + te_n$ , then

$$\|x\| \geq \delta|t|, \quad \|w\| \leq \|x\| + |t|\|e_n\| \leq (1 + \delta^{-1}\|e_n\|)\|x\|.$$

The first estimate bounds the last coordinate; the second, together with the inductive coordinate bounds on  $W$ , bounds all the preceding coordinates. Thus the coordinate map  $H \rightarrow K^n$ , with the maximum norm, is bounded. Its inverse is bounded by the ordinary triangle inequality. This gives a bi-Lipschitz linear identification with the complete space  $K^n$ , completing the induction. Proposition 15 now applies to whichever of  $E, E'$  is finite-dimensional.  $\square$

*Remark 17* (the coordinate hypothesis over an incomplete field). Finite algebraic dimension alone does not imply continuity of coordinate functionals over an incomplete field. For example, choose  $a \in \widehat{K} \setminus K$  and give  $H := K + Ka \subseteq \widehat{K}$  the induced norm. The vectors  $1, a$  form a  $K$ -basis. If  $a_j \in K$  tends to  $a$ , then  $a - a_j \rightarrow 0$  in  $H$ , whereas its  $a$ -coordinate is always 1. Thus that coordinate functional is discontinuous. This prevents dropping the coordinate hypothesis from the preceding proof; it does not by itself give a counterexample to analyticity of  $Q$  for such

spaces. Over an incomplete field the proposition still applies, for example, to finite-dimensional spaces whose norm is equivalent to a coordinate maximum norm.

Finite-dimensionality of the target  $F$  leads to a different question. Remark 50 reduces it to scalar-valued forms and gives a positive result over a spherically complete ultrametric base.

**4.3. Smoothness in every characteristic.** We first record why finite-order differentiability descends through a closed isometric inclusion. The argument also explains why smoothness survives when a multilinear analytic lift does not.

**Lemma 18** (smoothness through a closed isometric inclusion). *Let  $U, V, W$  be normed spaces over a nontrivially normed field  $K$ , and let  $j: V \rightarrow W$  be a linear isometry with closed range. For a map  $g: U \rightarrow V$  and  $n \in \mathbb{N} \cup \{\infty\}$ , the map  $g$  is  $C^n$  if and only if  $j \circ g$  is  $C^n$ .*

*Proof.* One direction follows by composition with a continuous linear map. For the converse, continuity follows from the isometry. Suppose  $j \circ g$  has Fréchet derivative  $A: U \rightarrow W$  at  $x$ . For each  $h \in U$ ,

$$Ah = \lim_{t \rightarrow 0, t \neq 0} \frac{j(g(x + th)) - j(g(x))}{t}.$$

The quotients lie in  $j(V)$ , and nontriviality of the norm supplies nonzero scalars tending to zero. Since  $j(V)$  is closed,  $Ah \in j(V)$ . Thus  $A = j \circ B$  for a unique linear map  $B: U \rightarrow V$ . The identity  $\|Bh\| = \|Ah\|$  makes  $B$  bounded, and the isometry transfers the Fréchet remainder estimate from  $A$  to  $B$ . Consequently  $g$  is differentiable and

$$D(j \circ g) = j_* \circ Dg, \quad j_*: \mathcal{L}(U, V) \longrightarrow \mathcal{L}(U, W), \quad T \longmapsto j \circ T.$$

The map  $j_*$  is again a linear isometry with closed range. Indeed, an operator belongs to its range exactly when all its values lie in  $j(V)$ ; this is a closed condition, being an intersection of inverse images under continuous evaluation maps. Applying the same argument to  $j_*$  proves the assertion for  $C^{n+1}$  from the assertion for  $C^n$ . Induction proves every finite order, and hence  $C^\infty$ .  $\square$

**Proposition 19** (smoothness of precomposition). *For every nontrivially normed field  $K$ , all normed  $K$ -spaces  $E, E', F$ , and every finite set  $\iota$ , the map  $Q_{E, E', F}^K$  is  $C^n$  for every  $n \in \mathbb{N} \cup \{\infty\}$ . In particular, precomposition is always  $C^\infty$ , without completeness, finite-dimensionality, or characteristic assumptions.*

*Proof.* Put  $k = |\iota|$  and use the finite-index identification of §3. The inclusion  $\text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$  is isometric and has closed range: alternation is the intersection of the closed conditions  $m(x_1, \dots, x_k) = 0$  for tuples with two equal entries. Postcomposition therefore gives a linear isometry with closed range

$$\mathcal{I}: \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)) \longrightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)).$$

In the larger target, precomposition is the diagonal of the bounded  $k$ -linear map

$$A(f_1, \dots, f_k)(m)(x_1, \dots, x_k) = m(f_1 x_1, \dots, f_k x_k),$$

whose norm is at most 1. Hence  $\mathcal{I} \circ Q$  is a continuous polynomial and is  $C^\infty$ . Lemma 18 gives the result. The degree-zero case is again constant.  $\square$

This is the argument formalized in [7]. It uses closedness to recover derivatives, rather than a projection onto alternating maps. The induction proves all finite orders but does not recover analytic coefficients in positive characteristic. Theorem 1 shows that the resulting  $C^\infty$  conclusion cannot in general be strengthened to  $C^\omega$ .

## 5. THE MULTIPLIER THEOREM OVER A FINITE FIELD

This section proves the algebraic obstruction used in the Laurent-series construction: over a finite field in which  $k! = 0$ , coordinatewise multiplication admits no multilinear lift satisfying a polarized diagonal identity and a uniform bound on exterior support. We first distinguish the diagonal identities, since over a finite field equality at all scalar points need not be a polynomial identity.

**5.1. Lifts and three diagonal identities.** Fix a field  $L$ , an integer  $k \geq 1$ ,  $L$ -vector spaces  $A$  and  $V$ , and a linear map  $D: A \rightarrow \text{End}_L(V)$ ,  $a \mapsto D_a$ . A *lift of  $D$  in degree  $k$*  is a  $2k$ -linear map  $\Psi: A^k \times V^k \rightarrow \Lambda^k V$  over  $L$  that is alternating in the  $V$ -slots; its diagonal identity will be specified separately. For every  $n \geq 1$  write  $[n] := \{1, \dots, n\}$ , and for  $\alpha \in \mathbb{N}^m$  put  $|\alpha| := \sum_i \alpha_i$ . Given  $m \geq 1$ , a map  $f: [k] \rightarrow [m]$ , a family  $b = (b_1, \dots, b_m) \in A^m$  and  $\lambda \in L^m$ , put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \text{type } f := (|f^{-1}(1)|, \dots, |f^{-1}(m)|) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^k \lambda_{f(j)} = \lambda^{\text{type } f}.$$

The *pointwise identity* is

$$(\text{Pw}) \quad \Psi(a, \dots, a; x_1, \dots, x_k) = D_a x_1 \wedge \dots \wedge D_a x_k \quad \text{for all } a \in A, x \in V^k.$$

The *polarized identity* is: for all  $m \geq 1$ ,  $b \in A^m$ ,  $\alpha \in \mathbb{N}^m$  with  $|\alpha| = k$ , and  $x \in V^k$ ,

$$(\text{Pol}) \quad \sum_{\text{type } f = \alpha} \Psi(b_f; x) = \sum_{\text{type } f = \alpha} D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k.$$

Its *multilinear instance* is: for all  $b_1, \dots, b_k \in A$  and  $x \in V^k$ ,

$$(\text{Pol1}) \quad \sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} D_{b_{\sigma(1)}} x_1 \wedge \dots \wedge D_{b_{\sigma(k)}} x_k.$$

Here  $\sum_{\text{type } f = \alpha}$  runs over the maps  $f: [k] \rightarrow [m]$  of type  $\alpha$ . [\(Pol1\)](#) is the instance  $m = k$ ,  $\alpha = (1, \dots, 1)$  of [\(Pol\)](#), since the maps of type  $(1, \dots, 1)$  are the permutations. For  $b \in A^m$  and  $x \in V^k$  put  $a_\lambda := \sum_i \lambda_i b_i$  and

$$\Delta_{b,x}(\lambda) := \Psi(a_\lambda, \dots, a_\lambda; x) - D_{a_\lambda} x_1 \wedge \dots \wedge D_{a_\lambda} x_k = \sum_{|\alpha|=k} \lambda^\alpha \delta_\alpha(b, x),$$

$$\delta_\alpha(b, x) := \sum_{\text{type } f = \alpha} \left[ \Psi(b_f; x) - D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k \right].$$

Multilinearity of  $\Psi$  in the  $A$ -slots, linearity of  $D$  and multilinearity of the wedge give the displayed expansion. Thus  $\Delta_{b,x}$  is a homogeneous polynomial of degree  $k$  with coefficients in  $\Lambda^k V$ . Identity [\(Pol\)](#) says that every coefficient  $\delta_\alpha(b, x)$  is zero. Identity [\(Pw\)](#) says instead that  $\Delta_{b,x}$  vanishes at every point of  $L^m$ ; conversely, this pointwise vanishing gives [\(Pw\)](#) by taking  $m = 1$  and  $\lambda = 1$ . The following two identity principles identify the cases in which pointwise vanishing forces all coefficients to vanish.

**Lemma 20** (identity principle). *Let  $\mathbb{K}$  be an infinite field,  $Y$  a  $\mathbb{K}$ -vector space and  $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$  a polynomial in  $\lambda \in \mathbb{K}^m$  with finitely many nonzero coefficients  $y_\alpha \in Y$ . If  $g(\lambda) = 0$  for all  $\lambda \in \mathbb{K}^m$ , then every  $y_\alpha = 0$ .*

*Proof.* The finitely many coefficients  $y_\alpha$  span a finite-dimensional subspace of  $Y$ . Applying its coordinate functionals reduces the claim to scalar polynomials. For these, induction on  $m$  reduces to the fact that a nonzero polynomial in one variable has only finitely many roots.  $\square$

**Lemma 21** (homogeneous forms over a finite field). *Let  $q$  be a prime power,  $L' \supseteq \mathbb{F}_q$  a field,  $1 \leq e \leq q$ , and  $g \in L'[\lambda_1, \dots, \lambda_m]$  homogeneous of degree  $e$  with  $g(\lambda) = 0$  for every  $\lambda \in \mathbb{F}_q^m$ . Then  $g = 0$ .*

*Proof.* Call a monomial  $\lambda^\beta$  *reduced* if every  $\beta_i \leq q-1$ . The indicator function of a point  $a \in \mathbb{F}_q^m$  is  $\prod_i (1 - (\lambda_i - a_i)^{q-1})$ , which expands into reduced monomials. So the  $q^m$  reduced monomials span the  $q^m$ -dimensional  $L'$ -space of functions  $\mathbb{F}_q^m \rightarrow L'$ , and they are linearly independent as functions. A monomial of degree  $e \leq q$  that is not reduced is  $\lambda_i^q$  (and then  $e = q \geq 2$ ); as a function on  $\mathbb{F}_q^m$  it agrees with the reduced monomial  $\lambda_i$ , of degree  $1 < e$ . If  $e = q$ , replace each  $\lambda_i^q$  in  $g$  by  $\lambda_i$ . These degree-one images are distinct, and homogeneity separates them from all the remaining monomials, which still have degree  $q$ . The replacement therefore sends the monomials of  $g$  injectively to reduced monomials without changing the associated function. Linear independence of the reduced monomials forces every coefficient to vanish. If  $e < q$ , no replacement is needed.  $\square$

**Proposition 22** (pointwise versus polarized). *Let  $\Psi$  be a lift of  $D$  in degree  $k$  over a field  $L$ .*

- (1)  $(\text{Pol}) \Rightarrow (\text{Pw}) \Rightarrow (\text{Poll})$ , over every field.
- (2) If  $L$  is infinite, or if  $L = \mathbb{F}_q$  and  $k \leq q$ , then  $(\text{Pw}) \Rightarrow (\text{Pol})$ .
- (3) If  $L = \mathbb{F}_q$  and  $k \geq q+1$ , then  $(\text{Pw})$  does not imply  $(\text{Pol})$  in general.
- (4) If  $k! = 0$  in  $L$ , then  $(\text{Poll})$  does not imply  $(\text{Pw})$  in general.

*Proof.* (1) *Inclusion-exclusion.* The implication  $(\text{Pol}) \Rightarrow (\text{Pw})$  is the case  $m = 1$  and  $\alpha = (k)$ . To prove  $(\text{Pw}) \Rightarrow (\text{Poll})$ , fix  $b_1, \dots, b_k \in A$  and  $x \in V^k$ , and put  $b_S := \sum_{i \in S} b_i$  for  $S \subseteq [k]$ . Multilinearity gives

$$\begin{aligned} \sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) &= \sum_S (-1)^{k-|S|} \sum_{f: [k] \rightarrow S} \Psi(b_f; x) \\ &= \sum_{f: [k] \rightarrow [k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x), \end{aligned}$$

because  $\sum_{S \supseteq T} (-1)^{k-|S|} = (1-1)^{k-|T|}$  is 1 if  $T = [k]$  and 0 otherwise. The same computation applies to the multilinear map  $(a_1, \dots, a_k) \mapsto D_{a_1} x_1 \wedge \dots \wedge D_{a_k} x_k$ , whose diagonal is the right side of  $(\text{Pw})$ . Evaluating  $(\text{Pw})$  at the  $2^k$  subset sums  $b_S$  therefore gives  $(\text{Poll})$ . The computation is valid in every characteristic and involves no division by  $k!$ .

(2) *The identity principles.* The coefficients of  $\Delta_{b,x}$  span a finite-dimensional subspace of  $\Lambda^k V$ ; in a basis of it each coordinate of  $\Delta_{b,x}$  is a scalar form of degree  $k$  vanishing on  $L^m$ . It is zero by Lemma 20 if  $L$  is infinite, and by Lemma 21 if  $L = \mathbb{F}_q$  and  $k \leq q$ .

(3) *Pointwise vanishing without full polarization.* Let  $\dim A \geq 3$ , with  $b_1, b_2, b_3 \in A$  and functionals  $g_1, g_2, g_3$  on  $A$  such that  $g_i(b_j) = \delta_{ij}$ , and let  $y_1, \dots, y_k \in V$  be linearly independent. Put

$$\theta(a_1, \dots, a_k) := [g_1(a_1) \cdots g_1(a_q) g_2(a_{q+1}) - g_1(a_1) g_2(a_2) \cdots g_2(a_{q+1})] \cdot g_3(a_{q+2}) \cdots g_3(a_k)$$

(for  $k = q+1$  the  $g_3$ -factor is empty and  $\dim A \geq 2$  suffices), and  $\Theta(a; x) := \theta(a) x_1 \wedge \dots \wedge x_k$ . Then  $\Theta$  is a lift of  $D = 0$ . It satisfies  $(\text{Pw})$ , because  $\theta(a, \dots, a) = (g_1(a)^q g_2(a) - g_1(a) g_2(a)^q) g_3(a)^{k-q-1} = 0$  since  $c^q = c$  for  $c \in \mathbb{F}_q$ . It violates  $(\text{Pol})$  for  $\alpha = (q, 1, k-q-1)$  and  $b = (b_1, b_2, b_3)$ : the first product contributes to  $\sum_{\text{type } f = \alpha} \theta(b_f)$  exactly for  $f = (1, \dots, 1, 2, 3, \dots, 3)$ , with value 1; the second product would need  $f(2) = \dots = f(q+1) = 2$ , that is, at least  $q \geq 2$  values equal to 2, which is impossible for type  $\alpha$ . So the coefficient  $\delta_\alpha(b, y)$  of  $\Theta$  is  $y_1 \wedge \dots \wedge y_k \neq 0$ . When  $k = q+1$ , the third label and its functional can be omitted throughout, with  $\alpha = (q, 1)$ .

(4) *The multilinear identity without the pointwise identity.* Take  $\bar{a} \in A$  and a functional  $g_1$  with  $g_1(\bar{a}) = 1$ , and  $y_1, \dots, y_k$  linearly independent. The lift  $\Theta'(a; x) := g_1(a_1) \cdots g_1(a_k) x_1 \wedge \cdots \wedge x_k$  of  $D = 0$  satisfies (Pol1), since  $\sum_{\sigma} \Theta'(b_{\sigma}; x) = k! g_1(b_1) \cdots g_1(b_k) x_1 \wedge \cdots \wedge x_k = 0$  for all  $b_1, \dots, b_k \in A$  and  $x \in V^k$ , but  $\Theta'(\bar{a}, \dots, \bar{a}; y) = y_1 \wedge \cdots \wedge y_k \neq 0$ .  $\square$

The finite-field obstruction below requires only (Pol1), which follows from either (Pw) or (Pol). The distinction between the latter two remains important: over  $\mathbb{F}_q$  they need not agree when  $k > q$ . The lift constructed in §6 nevertheless satisfies the full identity (Pol) in every degree: the identity is polarized over an infinite Laurent-series field before constant coefficients are taken (Lemma 32).

**5.2. Statement and proof strategy.** For the proof of the theorem, assume that  $L$  is a finite field with  $k! = 0$  in  $L$ , that is,  $\text{char } L = p \leq k$ ; in particular  $k \geq 2$ . Put  $V := L^{\mathbb{N}}$  and let  $V_{\text{fin}} := L^{(\mathbb{N})} \subseteq V$  be the subspace of finitely supported sequences. For  $u, x \in V$  we write  $ux$  for the coordinatewise product; so  $D_u x = ux$  is the multiplier family.

**Theorem 23** (multipliers over a finite field). *Let  $L$  be a finite field and  $k \geq 1$  with  $k! = 0$  in  $L$ .*

- (1) *There is no  $2k$ -linear map  $\Psi: V_{\text{fin}}^k \times V_{\text{fin}}^k \rightarrow \Lambda_L^k V$  over  $L$  such that*
  - (a)  *$\Psi$  is antisymmetric in the last  $k$  slots;*
  - (b)  *$\Psi$  satisfies (Pol1) for  $D_u x = ux$ : for all  $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\text{fin}}$ ,*

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k);$$

- (c) *for some  $d \in \mathbb{N}$ ,  $\text{sdim}(\Psi(u; x)) \leq d$  for all  $u, x \in V_{\text{fin}}^k$  with entries in  $\{0, \pm 1\}$ .*
- (2) *Let  $E_0 = V$  or  $E_0 = V_{\text{fin}}$ . There is no  $2k$ -linear map  $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_L^k E_0$  over  $L$  that is antisymmetric in the last  $k$  slots, satisfies (Pol1) for all finitely supported  $u, x$ , and has  $\text{sdim}(\Psi(u; x)) \leq d$  for all  $u, x \in E_0^k$ , for some  $d$ .*

*Reduction to (1).* Given a map as in (2), restrict its inputs to  $V_{\text{fin}}^k \times V_{\text{fin}}^k$  and compose its output with the injective map  $\Lambda^k E_0 \rightarrow \Lambda^k V$ . This preserves antisymmetry and (Pol1). Support dimension can only decrease under the inclusion of the ambient space (§3.3), so the resulting map satisfies (a)–(c).

We therefore prove (1). Suppose that  $\Psi$  satisfies its hypotheses, and fix the bound  $d$  in (c). Write  $e_i \in V_{\text{fin}}$  for the unit vectors,  $e_a := (e_{a_1}, \dots, e_{a_k})$  for  $a \in \mathbb{N}^k$ , and  $\Omega$  for the determinant array of §3.3 with  $S = \mathbb{N}$ .

The proof has three steps. Ramsey's theorem first makes the coefficients of  $\Omega\Psi$  depend only on the relative order and equalities among their indices. Special weights on four-point clusters then cancel every contribution from an occupied cluster containing no output index. This cancellation produces a two-valued staircase matrix from one exterior vector. Its bounded support forces the two values to agree, and hence makes all cluster order patterns have a common value. Finally, antisymmetry and (Pol1) say that the sum of these  $k!$  equal values is 1, contradicting  $k! = 0$  in  $L$ .

**5.3. Order-homogeneity from Ramsey's theorem.** We use Ramsey's theorem [8] in its infinite form: *for every  $n \geq 1$  and every coloring of the  $n$ -element subsets of  $\mathbb{N}$  with finitely many colors there is an infinite  $H \subseteq \mathbb{N}$  all of whose  $n$ -element subsets have the same color.*

For a tuple  $z \in \mathbb{N}^N$  with distinct values  $y_1 < \cdots < y_r$ , its *pattern* is the map  $i \mapsto j$  determined by  $z_i = y_j$ . In particular, repeated entries are retained in the pattern. Two tuples  $z$  and  $z'$  have the same pattern if and only if  $z_i < z_j \Leftrightarrow z'_i < z'_j$  and  $z_i = z_j \Leftrightarrow z'_i = z'_j$  for all  $i, j$ .

**Lemma 24** (pattern homogeneity). *For every  $N \geq 1$  and every function  $T: \mathbb{N}^N \rightarrow L$  there is an infinite  $H \subseteq \mathbb{N}$  such that  $T(z) = T(z')$  whenever  $z, z' \in H^N$  have the same pattern.*

*Proof.* Color an  $N$ -element set  $s = \{s_1 < \dots < s_N\} \subseteq \mathbb{N}$  by the vector

$$(T(s_{\beta(1)}, \dots, s_{\beta(N)}))_{\beta: [N] \rightarrow [N]} \in L^{N^N}.$$

The coloring records all maps  $\beta$ , including those that repeat indices. It has finitely many values because  $L$  is finite; this is the only use of the finiteness of  $L$  in the obstruction proof. Ramsey's theorem gives an infinite  $H$  on which the color is constant, say equal to  $(t_\beta)_\beta$ . Let  $z \in H^N$  have value set  $\{y_1 < \dots < y_r\}$ . Since  $H$  is unbounded, add  $N - r$  elements of  $H$  above  $y_r$  to obtain an  $N$ -element set  $s \subseteq H$ . Padding above the largest value leaves the ranks unchanged:  $s_j = y_j$  for  $j \leq r$ , and  $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$ , where  $\beta(i)$  is the rank of  $z_i$  in  $\{y_1, \dots, y_r\}$ . The map  $\beta$  depends only on the pattern of  $z$ , and  $T(z) = t_\beta$ .  $\square$

*Coefficients.* For  $(a; y; c) \in \mathbb{N}^k \times \mathbb{N}^k \times \mathbb{N}^k$  put

$$T(a; y; c) := \Omega_{\Psi(e_a; e_y)}(c_1, \dots, c_k) \in L.$$

Lemma 24 with  $N = 3k$  gives an infinite set  $H = \{h_0 < h_1 < h_2 < \dots\}$  on which  $T$  depends only on the pattern. For  $u_j, v_j \in V_{\text{fin}}$  supported in  $H$ , multilinearity of  $\Psi$  and linearity of  $\Omega$  give

$$(5.1) \quad \Omega_{\Psi(u; v)}(c) = \sum_{a, y \in H^k} \prod_{j=1}^k u_j(a_j) \prod_{j=1}^k v_j(y_j) \cdot T(a; y; c).$$

Although the sum is indexed by  $H^k \times H^k$ , only finitely many summands are nonzero, because all the input vectors have finite support. Every output tuple  $c$  used below also lies in  $H^k$ , so pattern homogeneity applies to all  $3k$  indices in (5.1).

**5.4. Clusters and the cluster value.** A *cluster* is a set  $C = \{h_n, h_{n+1}, h_{n+2}, h_{n+3}\}$  of four consecutive elements of  $H$ ; we write  $C(i) := h_{n+i-1}$  for  $i = 1, \dots, 4$ . Since a cluster is an interval of  $H$ , two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write  $C < C'$ . Distinct clusters need not be adjacent in  $H$ .

We choose two weight vectors, one for each kind of input slot:

$$\begin{aligned} s &= (s_1, s_2, s_3, s_4) := (1, 0, -1, 0) && \text{on the operator slots,} \\ w &= (w_1, w_2, w_3, w_4) := (1, -1, 1, -1) && \text{on the vector slots.} \end{aligned}$$

Over  $\mathbb{Z}$ , hence over every field,

$$(W) \quad \sum_i s_i = \sum_i w_i = \sum_i s_i w_i = \sum_{i < i'} s_i w_{i'} = \sum_{i > i'} s_i w_{i'} = 0, \quad s_1 w_1 = 1.$$

Indeed  $\sum_i s_i = 1 - 1$ ,  $\sum_i w_i = 1 - 1 + 1 - 1$ ,  $s_1 w_1 = 1$ , and

$$\begin{aligned} \sum_i s_i w_i &= s_1 w_1 + s_3 w_3 = 1 - 1 = 0, \\ \sum_{i < i'} s_i w_{i'} &= s_1(w_2 + w_3 + w_4) + s_3 w_4 = -1 + 1 = 0, \\ \sum_{i > i'} s_i w_{i'} &= s_3(w_1 + w_2) = -(1 - 1) = 0. \end{aligned}$$

The first two zero sums will cancel a cluster occupied by only one input index. The other three will cancel a cluster occupied by an operator index and a vector index, according to their relative order. For a cluster  $C$  put

$$s_C := \sum_{i=1}^4 s_i e_{C(i)}, \quad w_C := \sum_{i=1}^4 w_i e_{C(i)}.$$

These are vectors of  $V_{\text{fin}}$  with entries in  $\{0, \pm 1\}$ , and  $s_C w_C = e_{C(1)} - e_{C(3)}$ .

*The cluster value.* For pairwise disjoint clusters  $C_1, \dots, C_k$ , one for each slot, define

$$\begin{aligned} \chi(C_1, \dots, C_k) &:= \Omega_{\Psi(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})}(C_1(1), \dots, C_k(1)) \\ &= \sum_{i, i' \in [4]^k} \prod_{j=1}^k s_{i_j} w_{i'_j} \cdot T(C_1(i_1), \dots, C_k(i_k); C_1(i'_1), \dots, C_k(i'_k); C_1(1), \dots, C_k(1)). \end{aligned}$$

Let  $\tau \in S_k$  be the *order pattern* of  $(C_1, \dots, C_k)$ :  $\tau(j)$  is the position of  $C_j$  among  $C_1, \dots, C_k$ . The pattern of every  $3k$ -tuple in the sum is determined by  $(i, i')$  and  $\tau$ : entries in different clusters compare as the clusters do, and the entries  $C_j(i_j)$ ,  $C_j(i'_j)$ ,  $C_j(1)$  in the same cluster compare as their indices do, including equalities. Pattern homogeneity therefore shows that the displayed sum depends only on  $\tau$ . Every permutation occurs as the order pattern of some disjoint clusters in  $H$ ; we denote the resulting scalar by  $\chi(\tau)$ .

**Lemma 25** (cancellation of output-free clusters). *Let  $\mathcal{C}_1, \dots, \mathcal{C}_k$  be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in  $\bigcup_j \mathcal{C}_j$  are pairwise disjoint. Put  $u_j := \sum_{C \in \mathcal{C}_j} s_C$  and  $v_j := \sum_{C \in \mathcal{C}_j} w_C$ . Let  $C_l \in \mathcal{C}_l$  for  $l = 1, \dots, k$  and  $c := (C_1(1), \dots, C_k(1))$ . Then  $\Omega_{\Psi(u, v)}(c) = \chi(C_1, \dots, C_k)$ .*

*Proof.* We show that every cluster assignment except the one matching the output clusters has total contribution zero. Expand by (5.1). Since the clusters are disjoint, a nonzero term has, for each slot  $j$ , an operator index  $a_j = C_j^{\text{op}}(i_j)$  and a vector index  $y_j = C_j^{\text{vec}}(i'_j)$  with  $C_j^{\text{op}}, C_j^{\text{vec}} \in \mathcal{C}_j$ , and weight  $\prod_j s_{i_j} w_{i'_j}$ . Group the terms according to the *cluster assignment*  $(C_j^{\text{op}}, C_j^{\text{vec}})_j$ .

Call an occupied cluster *output-free* if it is none of  $C_1, \dots, C_k$ . Suppose a group has such a cluster  $\Gamma$ . Disjointness of the families assigns  $\Gamma$  to exactly one slot  $j$ , so the only tuple entries it can contain are  $a_j$  and  $y_j$ . Disjointness of the clusters excludes every output index from  $\Gamma$ .

Fix the positions of all indices outside  $\Gamma$  and sum over the positions inside it. Since  $\Gamma$  consists of consecutive elements of  $H$ , every outside entry lies either below the whole cluster or above it. Moving an index within  $\Gamma$  cannot change any comparison with an outside entry. By pattern homogeneity, the coefficient  $T$  therefore depends on the internal positions only as follows:

- (i) if only  $a_j \in \Gamma$ , not at all, and the partial sum is  $(\sum_i s_i) \cdot T = 0$ ;
- (ii) if only  $y_j \in \Gamma$ , not at all, and the partial sum is  $(\sum_i w_i) \cdot T = 0$ ;
- (iii) if both lie in  $\Gamma$ , only through whether  $i_j < i'_j$ ,  $i_j = i'_j$  or  $i_j > i'_j$ ; with the corresponding values  $T_<, T_=: T_>$  of  $T$  the partial sum is

$$T_< \sum_{i < i'} s_i w_{i'} + T_=: \sum_i s_i w_i + T_> \sum_{i > i'} s_i w_{i'} = 0.$$

In each case the weights of the other slots are a common factor, and the displayed partial sum vanishes by (W). Thus every assignment with an output-free cluster contributes zero. In any surviving assignment, every occupied cluster must be an output cluster. The only output cluster belonging to  $\mathcal{C}_j$  is  $C_j$ , which forces  $C_j^{\text{op}} = C_j^{\text{vec}} = C_j$  for every  $j$ . The contribution of this one assignment is exactly  $\chi(C_1, \dots, C_k)$ .  $\square$

**5.5. The staircase and the diagonal.** We now use the support bound to compare two order patterns that differ by an adjacent exchange. The key is to realize arbitrarily large staircase matrices as flattenings of a single value of  $\Psi$ .

**Lemma 26** (staircase). *Let  $\tau \in S_k$ , let  $\alpha \neq \beta$  be slots with  $\tau(\alpha) = \ell$  and  $\tau(\beta) = \ell + 1$ , and let  $\tau'$  be  $\tau$  with these two positions exchanged, so  $\tau'(\alpha) = \ell + 1$ ,  $\tau'(\beta) = \ell$  and  $\tau'(\gamma) = \tau(\gamma)$  otherwise. Then  $\chi(\tau) = \chi(\tau')$ .*

*Proof.* Let  $m \geq 1$ . Choose pairwise disjoint clusters, in increasing order in  $H$ : first a cluster  $C_\gamma$  for each slot  $\gamma$  with  $\tau(\gamma) < \ell$ , ordered as  $\tau$  prescribes; then  $2m$  clusters  $A_1 < B_1 < A_2 < B_2 < \dots < A_m < B_m$ ; then a cluster  $C_\gamma$  for each slot  $\gamma$  with  $\tau(\gamma) > \ell + 1$ , ordered as  $\tau$  prescribes. Put  $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$ ,  $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$  and  $\mathcal{C}_\gamma := \{C_\gamma\}$  for the other slots, and let  $u, v$  be as in Lemma 25 and  $\omega := \Psi(u; v)$ . Disjointness of the clusters ensures that the entries of  $u$  and  $v$  still lie in  $\{0, \pm 1\}$ , so  $\text{sdim}(\omega) \leq d$  by hypothesis (c) of Theorem 23(1). For this fixed  $m$ , keep  $u, v$  and hence  $\omega$  fixed while varying only the output indices.

Consider the  $m \times m$  matrix  $M$  with rows indexed by  $c_\alpha = A_n(1)$  and columns by  $c_\beta = B_{n'}(1)$  ( $n, n' = 1, \dots, m$ ), all other output indices being fixed at  $c_\gamma = C_\gamma(1)$ , and entries  $M_{nn'} := \Omega_\omega(c)$ . By Lemma 25, applied with the output clusters  $A_n \in \mathcal{C}_\alpha$ ,  $B_{n'} \in \mathcal{C}_\beta$  and  $C_\gamma \in \mathcal{C}_\gamma$ ,  $M_{nn'}$  is the cluster value of the tuple with  $A_n$  in slot  $\alpha$ ,  $B_{n'}$  in slot  $\beta$ , and  $C_\gamma$  in every other slot. The interlacing order gives  $A_n < B_{n'}$  exactly when  $n \leq n'$ . The tuple therefore has pattern  $\tau$  on and above the matrix diagonal, and pattern  $\tau'$  below it.

Put  $\delta := \chi(\tau) - \chi(\tau')$ , let  $\mathbf{1} \in L^m$  be the all-ones column, and let  $U$  be the upper unitriangular matrix with  $U_{nn'} = 1$  for  $n \leq n'$  and  $U_{nn'} = 0$  otherwise. Then

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau') \mathbf{1} \mathbf{1}^\top + \delta U,$$

The rows vary only slot  $\alpha$ ; each column fixes all the other slots, including the chosen value in slot  $\beta$ . Thus  $M$  is a submatrix of the flattening of this one array  $\Omega_\omega$  along slot  $\alpha$ . Lemma 12 gives  $\text{rank } M \leq \text{sdim}(\omega) \leq d$ .

If  $\delta \neq 0$ , the matrix  $\delta U$  is invertible, whereas the constant matrix  $\chi(\tau') \mathbf{1} \mathbf{1}^\top$  has rank at most one. Applying rank subadditivity to  $\delta U = M - \chi(\tau') \mathbf{1} \mathbf{1}^\top$  gives

$$m = \text{rank}(\delta U) \leq \text{rank } M + 1 \leq d + 1.$$

The construction is available for every  $m \geq 1$ ; choosing  $m = d + 2$  is a contradiction. Hence  $\delta = 0$ , as required.  $\square$

The support bound has now forced equality between adjacent order patterns. The remaining identity comes from (Poll); it does not use the support bound.

**Lemma 27** (the diagonal).  $\sum_{\tau \in S_k} \chi(\tau) = 1$ .

*Proof.* Take clusters  $C_1 < \dots < C_k$ , and put  $u_j := s_{C_j}$ ,  $v_j := w_{C_j}$  and  $c := (C_1(1), \dots, C_k(1))$ . Apply  $\omega \mapsto \Omega_\omega(c)$  to both sides of (Poll), with the operator vectors  $u_1, \dots, u_k$  and the vector arguments  $v_1, \dots, v_k$ .

*Right side.* If  $\sigma \neq \text{id}$ , some slot  $j$  has  $\sigma(j) \neq j$ . Disjoint supports then give  $u_{\sigma(j)} v_j = 0$ , so the corresponding wedge vanishes. Only  $\sigma = \text{id}$  survives. It gives  $\Omega_{(u_1 v_1) \wedge \dots \wedge (u_k v_k)}(c) = \det((u_b v_b)(C_a(1)))_{a,b}$ . Since  $u_b v_b = e_{C_b(1)} - e_{C_b(3)}$  and  $C_a(1) \neq C_b(3)$  for all  $a, b$ , this is the determinant of the identity matrix, so the right side is 1.

*Left side.* Fix  $\sigma \in S_k$  and write  $u_\sigma := (u_{\sigma(1)}, \dots, u_{\sigma(k)})$  and  $v_\sigma := (v_{\sigma(1)}, \dots, v_{\sigma(k)})$ . To identify the summand with a cluster value, we must permute both the vector arguments and the output coordinates to match the operator arguments. Each permutation contributes the same sign. More precisely, antisymmetry of  $\Psi$  in the vector slots gives  $\Psi(u_\sigma; v) = \text{sgn}(\sigma)\Psi(u_\sigma; v_\sigma)$ . For every  $\omega \in \Lambda^k V$  the function  $\Omega_\omega$  is alternating in its arguments, so  $\Omega_\omega(c) = \text{sgn}(\sigma)\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)})$ ; we apply this with  $\omega = \Psi(u_\sigma; v_\sigma)$ . Since  $\text{sgn}(\sigma)^2 = 1$ ,

$$\begin{aligned}\Omega_{\Psi(u_\sigma; v)}(c) &= \text{sgn}(\sigma)\Omega_{\Psi(u_\sigma; v_\sigma)}(c) = \text{sgn}(\sigma)^2\Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}).\end{aligned}$$

In the last expression slot  $j$  has operator  $s_{C_{\sigma(j)}}$ , vector  $w_{C_{\sigma(j)}}$  and output  $C_{\sigma(j)}(1)$ , so it equals  $\chi(C_{\sigma(1)}, \dots, C_{\sigma(k)})$  by definition. Its order pattern is  $\sigma$ , since  $C_{\sigma(j)}$  is the  $\sigma(j)$ -th cluster in increasing order. Thus the convention for patterns gives  $\sigma$ , not  $\sigma^{-1}$ . No cluster cancellation is needed in this calculation.

As  $\sigma$  runs over  $S_k$ , each order pattern occurs exactly once. Comparing the two sides gives

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma; v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1. \quad \square$$

*Proof of Theorem 23.* Exchanging adjacent ranks acts on an order pattern by  $\tau \mapsto (\ell \ell+1) \circ \tau$ ,  $1 \leq \ell < k$ . These transpositions generate  $S_k$ , and Lemma 26 makes  $\chi$  invariant under each exchange. Hence  $\chi$  is constant; write its value as  $\chi_0$ . Lemma 27 now gives  $k!\chi_0 = 1$  in  $L$ , contradicting  $k! = 0$ . This proves (1), and the preceding reduction gives (2).  $\square$

*Remark 28* (where the hypotheses enter). Ramsey homogeneity is the only step in the obstruction proof that uses the finiteness of  $L$ . Multilinearity then gives the cluster expansion and cancellation. The support bound is used in Lemma 26 only at the finitely many test inputs obtained by taking  $m = d + 2$ ; all have entries in  $\{0, \pm 1\}$ . Antisymmetry and (Poll) enter in Lemma 27, which does not use that bound. Finally,  $k! = 0$  is used only to turn the diagonal identity into a contradiction.

The inclusion–exclusion argument in Proposition 22(1) uses only multilinearity in the operator slots. The theorem therefore remains true with the pointwise identity (Pw) at  $L$ -points in place of (Poll), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma 27 are trivial.

*Remark 29* (when  $k!$  is invertible). The vanishing of  $k!$  is essential. If instead  $k! \neq 0$  in  $L$ , the *normalized alternatization*

$$\Psi_{\text{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)}x_1) \wedge \dots \wedge (u_{\sigma(k)}x_k)$$

is alternating in  $x$  and satisfies (Poll). Every summand is supported in the span of the  $k^2$  vectors  $u_i x_j$ , so  $\text{sdim}(\Psi_{\text{alt}}(u; x)) \leq k^2$ . The preceding homogeneity, cancellation, staircase and diagonal arguments apply, but now yield  $\chi \equiv 1/k!$ : for pairwise disjoint clusters only  $\sigma = \text{id}$  survives in  $\Psi_{\text{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$ , and the determinant at  $(C_1(1), \dots, C_k(1))$  is that of the identity. The final equation  $k! \cdot (1/k!) = 1$  is then consistent. So the obstruction is exactly the vanishing of  $k!$ , not positive characteristic as such. For example, at  $(p, k) = (5, 3)$  every step is consistent, since  $3! = 6 = 1$  in  $\mathbb{F}_5$  and  $\chi_0 = 1$ .

## 6. THE BANACH EXTERIOR TARGET OVER A LAURENT SERIES FIELD

We now realize the finite-field obstruction in a Banach-space setting. The target is the completion of an exterior power with its ordinary projective norm. Its determinant arrays

admit a constant-coefficient map into the *algebraic* exterior power over the coefficient field, with support dimension controlled by the Banach norm. This estimate turns a hypothetical bounded lift into the uniformly bounded-support map ruled out by Theorem 23. The passage to the completion is the key point: constant coefficients stabilize exactly once the approximation error is less than one.

**6.1. Setting.** Throughout this section  $k \geq 1$ ,  $\kappa$  is a field,  $r \in (0, 1)$ , and  $K_1 := \kappa((X))$  is the Laurent series field with absolute value  $|a| := r^{\text{ord } a}$  ( $|0| = 0$ ). This field is complete, nontrivially normed and ultrametric, and  $|c| = 1$  for every  $c \in \kappa^\times$ . Write  $\text{coeff}_n(a) \in \kappa$  for the  $X^n$ -coefficient of  $a \in K_1$ . For a set  $S$  and  $y \in \ell^\infty(S, K_1)$  we take the constant coefficient coordinatewise:

$$\text{coeff}_0(y) := (\text{coeff}_0(y(s)))_{s \in S} \in \kappa^S.$$

The resulting *constant-coefficient map*  $\text{coeff}_0$  is  $\kappa$ -linear but not  $K_1$ -linear. A nonzero Laurent series of absolute value less than one has order at least one. Consequently,

$$(6.1) \quad \|y\|_\infty < 1 \implies \text{coeff}_0(y) = 0, \quad \text{hence} \quad \|y - y'\|_\infty < 1 \implies \text{coeff}_0(y) = \text{coeff}_0(y').$$

Put  $E_1 := \ell^\infty(\mathbb{N}, K_1)$  with the sup norm, an ultrametric  $K_1$ -Banach space, and let  $E_0 := \kappa^\mathbb{N} \subseteq E_1$  be the  $\kappa$ -subspace of sequences with entries in  $\kappa$ . Every nonzero  $z \in E_0$  has  $\|z\|_\infty = 1$ . For  $x \in E_1$  and  $n \in \mathbb{Z}$  let  $x_{[n]} := (\text{coeff}_n(x(i)))_{i \in \mathbb{N}} \in E_0$  be the  $n$ -th Laurent component. For  $x \neq 0$  the set  $\{\text{ord } x(i) : x(i) \neq 0\}$  is bounded below, because  $r^{\text{ord } x(i)} \leq \|x\|_\infty$ , so it has a minimum  $\nu(x)$ ; then  $\|x\|_\infty = r^{\nu(x)}$  and  $x_{[n]} = 0$  for  $n < \nu(x)$ . Finally put

$$M_r := \max_{l \in \mathbb{N}} (l+1)r^l.$$

The maximum exists because  $(l+1)r^l \rightarrow 0$ . This constant will control the number of Laurent components that can contribute to a constant coefficient. For  $r \leq \frac{1}{2}$  one has  $(l+1)r^l \leq (l+1)2^{-l} \leq 1$ , so  $M_r = 1$ .

**6.2. The projective exterior norm.** On  $\Lambda := \Lambda_{K_1}^k E_1$  put

$$\|\omega\|_\pi := \inf \left\{ \sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty : \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k} \right\},$$

where the infimum is over finite decompositions and the cost is an ordinary sum. Let  $\Omega^{K_1} : \Lambda \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$  be the determinant array of §3.3 with  $L = K_1$ ,  $S = \mathbb{N}$  and  $V = E_1$ , so that  $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)(i_1, \dots, i_k) = \det(x_b(i_a))_{a,b}$ ; its values are bounded by the next lemma.

**Lemma 30.**  $\|\cdot\|_\pi$  is a  $K_1$ -norm on  $\Lambda$  with  $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$ , and  $\Omega^{K_1}$  is  $K_1$ -linear and injective with  $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$ . Let  $B$  be the completion of  $(\Lambda, \|\cdot\|_\pi)$ , a  $K_1$ -Banach space,  $J : B \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$  the continuous extension of  $\Omega^{K_1}$  (so  $\|J\| \leq 1$ ), and  $W_B : E_1^k \rightarrow B$ ,  $W_B(x) := x_1 \wedge \cdots \wedge x_k$ . Then  $W_B \in \text{Alt}_{K_1}^k(E_1; B)$ ,  $\|W_B\| \leq 1$ , and  $J(W_B(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$ .

*Proof.* For a pure wedge, each entry of the determinant array is a signed sum of products  $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$ . Each product has absolute value at most  $\prod_b \|x_b\|_\infty$ , so the ultrametric inequality in  $K_1$  gives the same bound for the determinant. For any finite decomposition of  $\omega$ , it follows that

$$\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty.$$

Thus the array is bounded, and taking the infimum gives  $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$ . The map  $\Omega^{K_1}$  is linear and injective by Lemma 11, with  $L = K_1$ ,  $S = \mathbb{N}$  and  $V = E_1$ . In particular,  $\|\omega\|_\pi > 0$  whenever  $\omega \neq 0$ .

The remaining norm laws follow directly from decompositions. Concatenation gives the triangle inequality. Since  $k \geq 1$ , a scalar can be absorbed into the first factor of each wedge; for  $\lambda \neq 0$  this gives  $\|\lambda\omega\|_\pi \leq |\lambda|\|\omega\|_\pi$ , and applying the same inequality to  $\lambda^{-1}$  gives equality. The pure-wedge bound follows by using a single summand.

The array space  $\ell^\infty(\mathbb{N}^k, K_1)$  is complete, so  $\Omega^{K_1}$  extends to the contraction  $J$  on  $B$ . The wedge map into  $B$  is  $K_1$ -multilinear and vanishes whenever two arguments coincide. The pure-wedge bound makes it continuous with norm at most one, and the asserted identity with  $J$  holds on its values by construction.  $\square$

For  $k \geq 2$ ,  $B$  is not ultrametric (Proposition 48).

**6.3. The support estimate.** We next show that the constant coefficient of every completed determinant array is represented by a finite algebraic exterior vector. The quantitative bound comes from counting the Laurent components needed as wedge *factors*; the number of wedge summands may be much larger.

Let  $\Omega^\kappa: \Lambda_\kappa^k E_0 \rightarrow \kappa^{\mathbb{N}^k}$  be the determinant array over  $\kappa$ ; it is injective by Lemma 11 with  $L = \kappa$  and  $V = E_0$ . For  $y_1, \dots, y_k \in E_0$  we have  $\Omega^{K_1}(y_1 \wedge \dots \wedge y_k) = \Omega^\kappa(y_1 \wedge \dots \wedge y_k)$ : the determinants are the same, with entries in  $\kappa \subseteq K_1$ . Support dimensions of elements of  $\Lambda_\kappa^k E_0$  are computed in  $E_0$  over  $\kappa$ .

**Proposition 31** (support estimate). *For every  $b \in B$  there is a unique  $\eta(b) \in \Lambda_\kappa^k E_0$  with  $\Omega^\kappa(\eta(b)) = \text{coeff}_0(Jb)$ . The map  $\eta$  is  $\kappa$ -linear, and*

$$(6.2) \quad \text{sdim}(\eta(b)) \leq kM_r \|b\|_B \quad \text{for all } b \in B.$$

*Proof. Step 1: one decomposable.* Let  $x_1, \dots, x_k \in E_1$ . If some  $x_b = 0$ , the wedge is zero and we take  $\eta = 0$ . Otherwise put  $\nu_b := \nu(x_b)$  and  $l := -\sum_b \nu_b$ , so that  $\prod_b \|x_b\|_\infty = r^{-l}$ . The possible exponent tuples contributing to a constant coefficient form the set

$$\mathcal{N} := \left\{ n \in \mathbb{Z}^k : n_b \geq \nu_b \text{ for all } b, \sum_b n_b = 0 \right\}.$$

For such a tuple, the nonnegative integers  $n_b - \nu_b$  sum to  $l$ . Hence  $\mathcal{N}$  is empty if  $l < 0$ ; if  $l \geq 0$ , every coordinate satisfies  $\nu_b \leq n_b \leq \nu_b + l$ , so  $\mathcal{N}$  is finite.

Fix  $(i_1, \dots, i_k) \in \mathbb{N}^k$  and expand each entry as  $x_b(i_a) = \sum_{n \geq \nu_b} \text{coeff}_n(x_b(i_a)) X^n$ . In the determinant expansion

$$\det(x_b(i_a))_{a,b} = \sum_\sigma \text{sgn}(\sigma) \prod_b x_b(i_{\sigma(b)}),$$

the constant coefficient of each product is its finite coefficient convolution over  $\mathcal{N}$ . Exchanging this finite sum with the permutation sum gives

$$\text{coeff}_0 \det(x_b(i_a))_{a,b} = \sum_{n \in \mathcal{N}} \det(x_{b,[n_b]}(i_a))_{a,b}.$$

The same finite set  $\mathcal{N}$  works for every array coordinate. Thus  $\text{coeff}_0(\Omega^{K_1}(x_1 \wedge \dots \wedge x_k)) = \Omega^\kappa(\eta)$ , where

$$\eta := \sum_{n \in \mathcal{N}} x_{1,[n_1]} \wedge \dots \wedge x_{k,[n_k]} \in \Lambda_\kappa^k E_0.$$

This is a finite sum in the algebraic exterior power. When  $l < 0$  it is zero. When  $l \geq 0$ , all its wedge factors belong to the set  $\{x_{b,[n]} : 1 \leq b \leq k, \nu_b \leq n \leq \nu_b + l\}$ , which has at most  $k(l+1)$  elements. Taking their span gives

$$\text{sdim}(\eta) \leq k(l+1) = k(l+1)r^l \cdot r^{-l} \leq kM_r \prod_b \|x_b\|_\infty.$$

The final bound therefore holds in both cases.

*Step 2: an element of  $\Lambda$ .* Write  $\omega = \sum_j w_j$ , where  $w_j = x_{j,1} \wedge \cdots \wedge x_{j,k}$ , and take  $\eta_j$  from Step 1. Additivity gives

$$\text{coeff}_0(\Omega^{K_1}(\omega)) = \sum_j \text{coeff}_0(\Omega^{K_1}(w_j)) = \Omega^\kappa\left(\sum_j \eta_j\right).$$

Injectivity of  $\Omega^\kappa$  shows that  $\eta(\omega) := \sum_j \eta_j$  is independent of the chosen decomposition. Subadditivity of support dimension now yields

$$\text{sdim}(\eta(\omega)) \leq \sum_j \text{sdim}(\eta_j) \leq kM_r \sum_j \prod_i \|x_{j,i}\|_\infty.$$

The left side is fixed while the decomposition varies. Taking the infimum therefore gives  $\text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_\pi$ , without requiring a minimizing decomposition.

*Step 3: completion.* Choose  $\omega_n \in \Lambda$  with  $\|b - \omega_n\|_B \rightarrow 0$ . Since  $J$  is a contraction,

$$\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \leq \|b - \omega_n\|_B < 1 \quad (n \geq n_0).$$

Equation (6.1) therefore gives the *exact* identities

$$\text{coeff}_0(Jb) = \text{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n)) \quad (n \geq n_0).$$

By injectivity of  $\Omega^\kappa$ , the algebraic vectors  $\eta(\omega_n)$  are eventually equal. Define  $\eta(b)$  to be this common value. Its array is  $\text{coeff}_0(Jb)$ , so the same injectivity also proves independence of the approximating sequence and uniqueness. In particular, no limit in  $\Lambda_\kappa^k E_0$  is involved.

For every  $n \geq n_0$  the fixed vector  $\eta(b)$  satisfies  $\text{sdim}(\eta(b)) \leq kM_r \|\omega_n\|_\pi$ . Since  $\|\omega_n\|_\pi \rightarrow \|b\|_B$ , this gives (6.2). Finally, the array characterization proves  $\kappa$ -linearity:  $J$  is  $K_1$ -linear,  $\text{coeff}_0$  is  $\kappa$ -linear, and  $\Omega^\kappa$  is linear and injective.  $\square$

**6.4. The coefficient lift.** The support estimate converts norm control into an algebraic obstruction. We apply it to a hypothetical lift, evaluated on coordinatewise multiplication operators and on the universal wedge map  $W_B$ .

Suppose that  $P$  is a bounded  $k$ -linear lift of  $Q_{E_1, E_1, B}^{K_1}$  over  $K_1$ . For  $a \in E_1$  let  $D_a \in \mathcal{L}(E_1, E_1)$  be coordinatewise multiplication,  $D_a x = ax$ . The map  $a \mapsto D_a$  is  $K_1$ -linear and  $\|D_a\| \leq \|a\|_\infty$ . Define

$$\begin{aligned} \Phi(a_1, \dots, a_k; x_1, \dots, x_k) &:= P(D_{a_1}, \dots, D_{a_k})(W_B)(x_1, \dots, x_k) \in B, \\ \Psi &:= \eta \circ \Phi|_{E_0^{2k}} : E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0. \end{aligned}$$

Since  $\Omega^\kappa \circ \eta = \text{coeff}_0 \circ J$ , the map  $\Psi$  is  $\text{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$ , read in  $\Lambda_\kappa^k E_0$  through the injective map  $\Omega^\kappa$ .

**Lemma 32.** *The map  $\Psi$  has the following properties.*

- ( $\Psi 1$ )  $\Psi$  is  $2k$ -linear over  $\kappa$ .
- ( $\Psi 2$ )  $\Psi$  is alternating in the last  $k$  slots.
- ( $\Psi 3$ )  $\text{sdim}(\Psi(z)) \leq d := \lfloor kM_r \|P\| \rfloor$  for all  $z \in E_0^{2k}$ .

(Ψ4)  $\Psi$  satisfies (Pol) for the multiplier family  $D^\kappa: E_0 \rightarrow \text{End}_\kappa(E_0)$ ,  $D_a^\kappa x = ax$ : for all  $m \geq 1$ ,  $b_1, \dots, b_m \in E_0$ ,  $\alpha \in \mathbb{N}^m$  with  $|\alpha| = k$ , and  $x \in E_0^k$ ,

$$\sum_{\text{type } f=\alpha} \Psi(b_f; x) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1) \wedge \cdots \wedge (b_{f(k)}x_k) \quad \text{in } \Lambda_\kappa^k E_0.$$

In particular  $\Psi$  satisfies (Poll).

*Proof.* The map  $\Phi$  is  $K_1$ -multilinear, so its restriction to  $E_0^{2k}$  is  $\kappa$ -multilinear. Composing with the  $\kappa$ -linear map  $\eta$  proves (Ψ1). Since  $P(D_{a_1}, \dots, D_{a_k})(W_B)$  belongs to  $\text{Alt}_{K_1}^k(E_1; B)$ , it vanishes when two of the last  $k$  arguments coincide; the same holds after applying  $\eta$ , proving (Ψ2).

For (Ψ3), all inputs from  $E_0$  have norm at most one. Thus

$$\|\Phi(a; x)\| \leq \|P\| \prod_i \|D_{a_i}\| \|W_B\| \prod_i \|x_i\|_\infty \leq \|P\|.$$

Apply (6.2) and use that support dimension is an integer to obtain the stated bound by  $d$ .

To prove (Ψ4), we first compare coefficients over the infinite field  $K_1$ , while the values still lie in  $B$ . Fix  $b_1, \dots, b_m \in E_0$  and  $x \in E_0^k$ . For  $\lambda \in K_1^m$  put  $a_\lambda := \sum_i \lambda_i b_i \in E_1$ . The diagonal identity for the lift gives

$$\Phi(a_\lambda, \dots, a_\lambda; x) = P(D_{a_\lambda}, \dots, D_{a_\lambda})(W_B)(x) = Q(D_{a_\lambda})(W_B)(x) = W_B(a_\lambda x_1, \dots, a_\lambda x_k).$$

Expanding both sides by multilinearity gives  $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$  for all  $\lambda \in K_1^m$ , where

$$L_\alpha := \sum_{\text{type } f=\alpha} \Phi(b_f; x) \in B, \quad R_\alpha := \sum_{\text{type } f=\alpha} W_B(b_{f(1)}x_1, \dots, b_{f(k)}x_k) \in B.$$

The field  $K_1$  is infinite even when  $\kappa$  is finite. Lemma 20, applied in the  $K_1$ -vector space  $B$ , therefore yields  $L_\alpha = R_\alpha$  for every  $\alpha$ .

We can now apply  $\eta$ . Additivity gives  $\eta(L_\alpha) = \sum_{\text{type } f=\alpha} \Psi(b_f; x)$ . To identify the other side, take  $y_1, \dots, y_k \in E_0$ . By Lemma 30 and the determinant-array agreement preceding Proposition 31,

$$J(W_B(y)) = \Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^\kappa(y_1 \wedge \cdots \wedge y_k).$$

This array has entries in  $\kappa$ , so taking its constant coefficient leaves it unchanged. Uniqueness in Proposition 31 then gives  $\eta(W_B(y)) = y_1 \wedge \cdots \wedge y_k \in \Lambda_\kappa^k E_0$ . Using  $y_j = b_{f(j)}x_j$  in each summand of  $R_\alpha$  proves

$$\eta(R_\alpha) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1) \wedge \cdots \wedge (b_{f(k)}x_k),$$

which is the required identity.  $\square$

The order of these operations is essential for the full identity (Ψ4). Evaluating the diagonal only at  $\kappa$ -points would give (Pw) there, which for  $\kappa = \mathbb{F}_q$  and  $k > q$  is strictly weaker than (Pol) (Proposition 22(3)). The proof below uses only the consequence (Poll).

### 6.5. No bounded lift over a Laurent series field.

**Theorem 33.** *Let  $\kappa$  be a finite field of characteristic  $p$ , let  $k \geq p$  and  $r \in (0, 1)$ , and let  $K_1 = \kappa((X))$  with  $|X| = r$ . Let  $E_1 = \ell^\infty(\mathbb{N}, K_1)$ , and let  $B$  be the completion of  $\Lambda_{K_1}^k E_1$  under the projective exterior norm. Then  $Q_{E_1, E_1, B}^{K_1}$  admits no bounded  $k$ -linear lift over  $K_1$ . Consequently it is analytic at no point of  $\mathcal{L}(E_1, E_1)$ .*

*Proof.* Suppose  $P$  exists, and form  $\Psi$  as in §6.4. By Lemma 32,  $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0$  with  $E_0 = \kappa^\mathbb{N}$  is  $2k$ -linear over  $\kappa$ , alternating and hence antisymmetric in the last  $k$  slots, satisfies (Pol1), and has  $\text{sdim}(\Psi(z)) \leq d$  for all inputs  $z$ . Moreover  $k! = 0$  in  $\kappa$ , since  $k \geq p = \text{char } \kappa$ . This contradicts Theorem 23(2) with  $L = \kappa$  and  $E_0 = \kappa^\mathbb{N}$ . The last sentence follows from Proposition 8.  $\square$

*Remark 34.* Lemmas 30 and 32 and Proposition 31 hold for every field  $\kappa$ , and nothing in them uses a normalization of  $r$ : the constant  $M_r$  makes the argument uniform in  $r \in (0, 1)$ . Only the final appeal to Theorem 23 needs  $\kappa$  finite. We do not need, and do not discuss, the analogue of Theorem 33 for infinite  $\kappa$  with this particular pair  $(E_1, B)$ . For an infinite  $\kappa$  of characteristic  $p$  the field  $\kappa((X))$  is covered by Theorem 1, with the pair obtained by ascent from its subfield  $\mathbb{F}_p((X))$ .

## 7. ASCENT TO AN ARBITRARY BASE FIELD

To pass from a Laurent series field to a prescribed field, we work first inside its completion. A suitable Laurent subfield is spherically complete, so Ingleton's theorem supplies a contractive scalar retraction. The ordinary projective tensor norm then gives isometric scalar extensions of the source and target, together with a retraction onto the original Banach target. Operators and alternating forms extend compatibly with these embeddings. As a result, any lift after scalar extension descends to the Laurent subfield, and nonexistence of a lift ascends. The final density argument removes the completeness assumption on the prescribed field.

**7.1. The Laurent subfield.** We use spherical completeness in the sense of §2: every nonempty family of pairwise-meeting closed balls has a common point.

**Lemma 35** (discrete distances). *Let  $Y$  be a complete ultrametric space whose distances lie in  $r^\mathbb{Z} \cup \{0\}$  for some  $r \in (0, 1)$ . Then  $Y$  is spherically complete.*

*Proof.* Write  $\mathcal{D}(a, \rho) = \{y \in Y : d(y, a) \leq \rho\}$ . In an ultrametric space, every point of a ball is a center. Consequently, whenever two closed balls meet, the one with the smaller radius lies in the other. The discrete distance assumption also gives

$$\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^b), \quad \rho^b := \max\{s \in r^\mathbb{Z} \cup \{0\} : s \leq \rho\}.$$

We may therefore replace every radius by a value in  $r^\mathbb{Z} \cup \{0\}$ .

Let  $(\mathcal{D}(a_i, \rho_i))_{i \in I}$  be a nonempty family of pairwise meeting closed balls. If its radii have a least element  $\rho_{i_0}$ , then  $\mathcal{D}(a_{i_0}, \rho_{i_0})$  lies in every ball of the family, so  $a_{i_0}$  is a common point. Otherwise, discreteness implies that all the radii are positive and that their infimum is zero. Choose  $i_1, i_2, \dots$  so that  $\rho_{i_n}$  decreases strictly to zero. The corresponding balls are nested, and

$$d(a_{i_{n'}}, a_{i_n}) \leq \rho_{i_n} \quad (n' \geq n).$$

Their centers form a Cauchy sequence. By completeness it has a limit  $a$ , and closedness puts  $a$  in every ball of the sequence. For any  $i \in I$ , choose  $n$  with  $\rho_{i_n} \leq \rho_i$ . The  $n$ -th ball then lies in  $\mathcal{D}(a_i, \rho_i)$ , so  $a$  belongs to every ball of the original family.  $\square$

*Remark 36* (the hypothesis (K2) of [6]). The construction of [6] assumes that the base field satisfies the hypothesis (K2) there: there are closed balls  $\mathcal{D}(c_n, r_n)$  with  $|c_{n+1} - c_n| \leq r_n$ ,  $r_0 > r_1 > \dots$  and  $\bigcap_n \mathcal{D}(c_n, r_n) = \emptyset$ . An ultrametric field (indeed any ultrametric space) satisfies (K2) if and only if it is not spherically complete. Indeed, the balls in (K2) pairwise meet, since  $|c_n - c_m| \leq \max_{m \leq j < n} |c_{j+1} - c_j| \leq r_m$  for  $m \leq n$ . Conversely, let a family of pairwise meeting closed balls have no common point. A ball of least radius in the family would lie in all the others (as in the proof of Lemma 35), and its center would be a common point;

so the radii have no least element, and every radius in the family exceeds their infimum. A sequence of balls from the family whose radii decrease strictly to that infimum is then nested, so it satisfies  $|c_{n+1} - c_n| \leq r_n$ , and it has empty intersection: a common point of the sequence would lie in every ball of the family, because every ball of the family contains a ball of the sequence. This remark is used only in §1 and §10.

**Lemma 37** (Laurent subfield). *Let  $K'$  be a complete nontrivially normed field of characteristic  $p$ , and let  $t \in K'$  with  $0 < |t| = r < 1$ .*

- (1)  $K'$  is ultrametric, and  $|c| = 1$  for every  $c \in \mathbb{F}_p^\times$ .
- (2) For  $a_m, \dots, a_M \in \mathbb{F}_p$ , not all zero,  $|\sum_{i=m}^M a_i t^i| = r^{\min\{i: a_i \neq 0\}}$ .
- (3) Give  $\mathbb{F}_p((X))$  the absolute value with  $|X| = r$ . There is a continuous isometric ring homomorphism  $\text{ev}_t: \mathbb{F}_p((X)) \rightarrow K'$  such that  $\text{ev}_t(\sum_{i \geq m} a_i X^i) = \sum_{i \geq m} a_i t^i$ . Its image  $K_1$  is a closed subfield of  $K'$ ; it is the closure of  $\mathbb{F}_p(t)$ , and it is isometrically isomorphic to  $\mathbb{F}_p((X))$ .
- (4)  $K_1$  is spherically complete.

*Proof.* (1) Lemma 6 makes  $K'$  ultrametric. For  $c \in \mathbb{F}_p^\times$ , the identity  $c^{p-1} = 1$  gives  $|c| = 1$ .

(2) The nonzero terms  $a_i t^i$  have pairwise distinct absolute values  $r^i$ . In an ultrametric field,  $|x + y| = |x|$  whenever  $|y| < |x|$ . Repeatedly applying this identity shows that the sum has the absolute value of its largest term, namely  $r^{\min\{i: a_i \neq 0\}}$ .

(3) Evaluation on Laurent polynomials defines a ring homomorphism  $\text{ev}_t: \mathbb{F}_p[X, X^{-1}] \rightarrow K'$ , and (2) shows that  $|\text{ev}_t(g)| = r^{\text{ord } g} = |g|$ . Laurent polynomials are dense in  $\mathbb{F}_p((X))$ , since truncations converge. Completeness of  $K'$  therefore gives a unique continuous extension to  $\mathbb{F}_p((X))$ . Passing to limits preserves the isometry and the ring operations. It also gives the stated evaluation formula, because the partial sums of  $\sum_{i \geq m} a_i X^i$  map to the partial sums of  $\sum_{i \geq m} a_i t^i$ .

The image is complete as the isometric image of a complete space, hence closed in  $K'$ . It is a subfield: for  $g \neq 0$ , the identity  $\text{ev}_t(g)\text{ev}_t(g^{-1}) = 1$  provides the inverse. This subfield contains  $\mathbb{F}_p(t)$ , while the Laurent polynomials  $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$  are dense in it. The image is consequently exactly the closure of  $\mathbb{F}_p(t)$ .

(4) By (3),  $K_1$  is complete and has value group  $|K_1^\times| = r^\mathbb{Z}$ . It is ultrametric by (1), so its distances lie in  $r^\mathbb{Z} \cup \{0\}$ . Lemma 35 gives spherical completeness.  $\square$

**7.2. Ingleton's theorem and its consequences.** Throughout the rest of this section we fix fields  $K_1 \subseteq K'$  with the following properties.

- (H1)  $K_1$ , with the restricted absolute value, is nontrivially normed, complete and spherically complete.
- (H2)  $K'$  is ultrametric.

By multiplication,  $K'$  is an ultrametric normed  $K_1$ -space. In our application  $K' = \widehat{K}$  and  $K_1$  is the field of Lemma 37.

**Lemma 38.** *Let  $\mathbb{K}$  be a complete nontrivially normed field and  $U$  a finite-dimensional normed  $\mathbb{K}$ -space. Then every linear functional on  $U$  is bounded, and  $U$  is complete.*

*Proof.* We prove both assertions by induction on  $n = \dim U$ , the case  $n = 0$  being immediate. Suppose  $n \geq 1$  and let  $g \neq 0$  be a linear functional on  $U$ . Its kernel  $H$  has dimension  $n - 1$ , so the induction hypothesis makes  $H$  complete and therefore closed in  $U$ . Choose  $v$  with  $g(v) = 1$ . Since  $v \notin H$ ,  $\delta := \inf_{h \in H} \|v - h\| > 0$ . Write  $x = g(x)v + h$  with  $h \in H$ . If  $g(x) \neq 0$ , then

$$\|x\| = |g(x)| \|v + g(x)^{-1}h\| \geq |g(x)|\delta.$$

Thus  $|g(x)| \leq \delta^{-1}\|x\|$  for every  $x$ , proving boundedness.

Choose a basis of  $U$ . Its coordinate functionals are now bounded, so the coordinate map  $U \rightarrow \mathbb{K}^n$ , with the max norm on  $\mathbb{K}^n$ , is bounded. The inverse map  $(c_i) \mapsto \sum_i c_i e_i$  is bounded by  $\sum_i \|e_i\|$ . Hence  $U$  is bi-Lipschitz equivalent to the complete space  $\mathbb{K}^n$ , and is itself complete.  $\square$

**Theorem 39** (Ingleton [3]; see also [10]). *Let  $K_1$  be a spherically complete ultrametric normed field,  $V$  a  $K_1$ -vector space with an ultrametric norm,  $V_0 \subseteq V$  a subspace,  $C \geq 0$ , and  $T_0: V_0 \rightarrow K_1$  linear with  $|T_0 v| \leq C\|v\|$  for  $v \in V_0$ . Then  $T_0$  extends to a linear map  $T: V \rightarrow K_1$  with  $|Tx| \leq C\|x\|$  for all  $x \in V$ .*

We will use Theorem 39 on the ultrametric  $K_1$ -space  $K'$  in two ways: to obtain the scalar retraction below, and to extend finite-dimensional coordinate functionals in the proof of positivity of the projective tensor norm. Both applications have codomain  $K_1$ . The spaces  $E_1$  and  $F_1$  introduced later need not be ultrametric.

**Corollary 40** (Ingleton projection). *There is a  $K_1$ -linear map  $\varpi: K' \rightarrow K_1$  with  $\varpi|_{K_1} = \text{id}$  and  $|\varpi(\lambda)| \leq |\lambda|$  for all  $\lambda \in K'$ .*

*Proof.* Theorem 39 with  $V = K'$ ,  $V_0 = K_1$ ,  $T_0 = \text{id}$  and  $C = 1$ ; its hypotheses hold by (H1) and (H2).  $\square$

**Lemma 41** (extension). *Let  $\mathbb{K} \in \{K_1, K'\}$  and let  $C \geq 0$ .*

- (1) *Let  $V_0$  be a dense  $\mathbb{K}$ -subspace of a normed  $\mathbb{K}$ -space  $V$ , let  $Y$  be a  $\mathbb{K}$ -Banach space and  $T_0: V_0 \rightarrow Y$  a  $\mathbb{K}$ -linear map with  $\|T_0 x\| \leq C\|x\|$ . Then  $T_0$  extends uniquely to a continuous map  $T: V \rightarrow Y$ , which is  $\mathbb{K}$ -linear with  $\|Tx\| \leq C\|x\|$ .*
- (2) *Let  $V_1, \dots, V_k$  be normed  $K'$ -spaces with dense subspaces  $V_i^0$ ,  $Y$  a  $K'$ -Banach space, and  $M: \prod_i V_i^0 \rightarrow Y$  a  $K'$ -multilinear map with  $\|M(d)\| \leq C \prod_i \|d_i\|$ . Then  $M$  extends uniquely to a continuous  $K'$ -multilinear map  $\bar{M}: \prod_i V_i \rightarrow Y$  with  $\|\bar{M}\| \leq C$ .*

*Proof.* (1) The bound makes  $T_0$  Lipschitz, hence uniformly continuous. Density of  $V_0$  and completeness of  $Y$  give a unique continuous extension. Linearity and the norm bound pass to limits by continuity of addition, scalar multiplication and the norm.

(2) The case  $k = 0$  is immediate. For  $k \geq 1$ , multilinearity gives the telescoping estimate

$$\|M(d) - M(d')\| \leq C\rho^{k-1} \sum_i \|d_i - d'_i\|$$

whenever  $\|d_i\|, \|d'_i\| \leq \rho$  for every  $i$ , with  $\rho > 0$ . For any  $v \in \prod_i V_i$ , choose tuples  $d^{(n)} \in \prod_i V_i^0$  converging to  $v$ . Their coordinates are uniformly bounded, so the estimate makes  $M(d^{(n)})$  Cauchy in  $Y$ . Define  $\bar{M}(v)$  to be its limit. The same estimate, applied to two approximating sequences, proves that this definition is independent of the choices and is continuous on bounded sets. Multilinearity and the product norm bound pass to limits. Finally, density of  $\prod_i V_i^0$  proves uniqueness.  $\square$

**7.3. Projective base change.** The next construction extends scalars while retaining the original norm and a contractive retraction whenever the original space is complete. The scalar retraction from Corollary 40 is the essential ingredient; the normed spaces themselves may have ordinary, non-ultrametric norms.

For a normed  $K_1$ -space  $V$  let  $V_{K'} := V \otimes_{K_1} K'$ , a  $K'$ -vector space through the second factor, and put

$$\|u\|_\pi := \inf \left\{ \sum_j \|x_j\| |\lambda_j| : u = \sum_j x_j \otimes \lambda_j \right\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded  $K_1$ -linear  $\psi: K' \rightarrow K_1$  let  $T_\psi: V_{K'} \rightarrow V$  be the  $K_1$ -linear map with  $T_\psi(x \otimes \lambda) = \psi(\lambda)x$ ; it exists because  $(x, \lambda) \mapsto \psi(\lambda)x$  is  $K_1$ -bilinear. Let  $\iota_V: V \rightarrow V_{K'}$ ,  $x \mapsto x \otimes 1$ ; it is  $K_1$ -linear, since  $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$  for  $c \in K_1$ .

**Lemma 42** (projective base change). *Let  $V$  be a normed  $K_1$ -space; no ultrametric hypothesis is made on  $V$ .*

- (1)  $\|T_\psi u\| \leq \|\psi\| \|u\|_\pi$  for every bounded  $K_1$ -linear  $\psi: K' \rightarrow K_1$ .
- (2)  $\|\cdot\|_\pi$  is a  $K'$ -norm on  $V_{K'}$ .
- (3)  $\iota_V$  is isometric.
- (4) Let  $V \widehat{\otimes}_\pi K'$  be the completion of  $(V_{K'}, \|\cdot\|_\pi)$ . If  $V$  is complete, then  $T_\varpi$ , with  $\varpi$  as in Corollary 40, extends to a  $K_1$ -linear map  $\Pi_V: V \widehat{\otimes}_\pi K' \rightarrow V$  with  $\|\Pi_V\| \leq 1$  and  $\Pi_V \circ \iota_V = \text{id}_V$ .

*Proof.* (1) For every decomposition  $u = \sum_j x_j \otimes \lambda_j$ ,

$$\|T_\psi u\| = \left\| \sum_j \psi(\lambda_j) x_j \right\| \leq \sum_j \|\psi\| |\lambda_j| \|x_j\|.$$

Taking the infimum over decompositions proves the contraction estimate.

(2) Concatenating decompositions gives the triangle inequality. Multiplying their second factors by  $\mu \in K'$  gives  $\|\mu u\|_\pi \leq |\mu| \|u\|_\pi$ ; for  $\mu \neq 0$ , applying the same inequality with  $\mu^{-1}$  gives equality. The zero-scalar case is immediate.

It remains to prove positivity. Let  $u \neq 0$ . Choose a finite tensor representation and rewrite its second factors in a basis of their  $K_1$ -span. This gives

$$u = \sum_{j=1}^n x_j \otimes \lambda_j$$

with  $\lambda_1, \dots, \lambda_n$  linearly independent over  $K_1$ . Some  $x_{j_0}$  is nonzero, since otherwise  $u = 0$ . On the finite-dimensional subspace  $U := \text{span}_{K_1}\{\lambda_1, \dots, \lambda_n\} \subseteq K'$ , take the coordinate functional  $\psi_0(\lambda_j) = \delta_{jj_0}$ . It is bounded by some  $C \geq 0$  by Lemma 38, using completeness of  $K_1$  in (H1). Apply Theorem 39 to the ultrametric  $K_1$ -space  $K'$  to extend it to  $\psi: K' \rightarrow K_1$  with  $\|\psi\| \leq C$ . Then  $T_\psi u = x_{j_0}$ , so (1) gives

$$0 < \|x_{j_0}\| \leq C \|u\|_\pi.$$

Thus  $\|u\|_\pi > 0$ . This argument detects a nonzero vector coefficient by a functional on the *scalar* factor  $K'$ ; it requires no separating continuous dual on  $V$ .

(3) The elementary tensor representation gives  $\|x \otimes 1\|_\pi \leq \|x\|$ . For the reverse inequality, use  $\varpi(1) = 1$  and  $\|\varpi\| \leq 1$  in (1):

$$\|x\| = \|T_\varpi(x \otimes 1)\| \leq \|x \otimes 1\|_\pi.$$

(4) When  $V$  is complete, the contraction  $T_\varpi$  on the dense tensor subspace extends by Lemma 41(1), with  $\mathbb{K} = K_1$ , to a  $K_1$ -linear contraction  $\Pi_V$  with values in  $V$ . On the embedded copy of  $V$  it satisfies  $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$ .  $\square$

We now fix the spaces to which the descent argument will be applied. Let  $E_1$  be a normed  $K_1$ -space and  $F_1$  a  $K_1$ -Banach space, and put

$$E := E_1 \widehat{\otimes}_\pi K', \quad F := F_1 \widehat{\otimes}_\pi K',$$

the completions of  $((E_1)_{K'}, \|\cdot\|_\pi)$  and  $((F_1)_{K'}, \|\cdot\|_\pi)$ . They are  $K'$ -Banach spaces;  $\iota_E: E_1 \rightarrow E$  and  $\iota_F: F_1 \rightarrow F$  are  $K_1$ -linear isometries, and  $\Pi_F: F \rightarrow F_1$  is  $K_1$ -linear with  $\|\Pi_F\| \leq 1$  and  $\Pi_F \circ \iota_F = \text{id}_{F_1}$  (Lemma 42).

**Lemma 43** (operators). *For  $f \in \mathcal{L}_{K_1}(E_1, E_1)$  the map  $f \otimes \text{id}$  on  $(E_1)_{K'}$  is  $K'$ -linear with  $\|(f \otimes \text{id})u\|_\pi \leq \|f\| \|u\|_\pi$ . Its extension  $f_{K'} \in \mathcal{L}_{K'}(E, E)$  satisfies  $\|f_{K'}\| \leq \|f\|$  and  $f_{K'} \circ \iota_E = \iota_E \circ f$ , and  $f \mapsto f_{K'}$  is  $K_1$ -linear.*

*Proof.* For any decomposition  $u = \sum_j x_j \otimes \lambda_j$ , the image under  $f \otimes \text{id}$  has the decomposition  $\sum_j f(x_j) \otimes \lambda_j$ , whose cost is at most  $\|f\| \sum_j \|x_j\| \|\lambda_j\|$ . Taking the infimum proves the norm bound on the algebraic tensor product. Since  $E$  is complete, Lemma 41(1), with  $\mathbb{K} = K'$ , gives the continuous extension  $f_{K'}$  with the same bound. Its action on  $x \otimes 1$  proves the embedding identity. Additivity and  $K_1$ -homogeneity in  $f$  hold on elementary tensors, hence on their span, and then on  $E$  by continuity.  $\square$

**Lemma 44** (forms). *For  $m \in \text{Alt}_{K_1}^k(E_1; F_1)$  there is  $m_{K'} \in \text{Alt}_{K'}^k(E; F)$  with  $\|m_{K'}\| \leq \|m\|$  and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F(m(x_1, \dots, x_k)) \quad (x_i \in E_1),$$

*and  $m \mapsto m_{K'}$  is  $K_1$ -linear.*

*Proof.* We first define the form on algebraic tensors, then establish a bound independent of their decompositions, and finally pass to the completions. Fix a  $K_1$ -basis  $(\lambda_\beta)_{\beta \in \mathfrak{B}}$  of  $K'$ . Every  $u \in (E_1)_{K'}$  has a unique finite expansion  $u = \sum_\beta u^\beta \otimes \lambda_\beta$ , with  $u^\beta \in E_1$ , and each coordinate map  $u \mapsto u^\beta$  is  $K_1$ -linear. Define

$$m^{\text{alg}}(u_1, \dots, u_k) := \sum_{\beta_1, \dots, \beta_k} m(u_1^{\beta_1}, \dots, u_k^{\beta_k}) \otimes \lambda_{\beta_1} \cdots \lambda_{\beta_k} \in (F_1)_{K'}.$$

The sum is finite, since each input has finitely many nonzero coordinates.

- (i) Linearity of the coordinate maps and multilinearity of  $m$  make  $m^{\text{alg}}$   $K_1$ -multilinear.
- (ii) On elementary tensors, the definition gives

$$m^{\text{alg}}(x_1 \otimes \mu_1, \dots, x_k \otimes \mu_k) = m(x_1, \dots, x_k) \otimes \mu_1 \cdots \mu_k \quad (x_i \in E_1, \mu_i \in K').$$

Indeed, writing  $\mu_i = \sum_\beta c_{i\beta} \lambda_\beta$  with  $c_{i\beta} \in K_1$  gives  $x_i \otimes \mu_i = \sum_\beta (c_{i\beta} x_i) \otimes \lambda_\beta$ . Substitution in the defining sum yields

$$\sum_{\beta_1, \dots, \beta_k} \left( \prod_i c_{i\beta_i} \right) m(x) \otimes \prod_i \lambda_{\beta_i} = m(x) \otimes \prod_i \left( \sum_\beta c_{i\beta} \lambda_\beta \right),$$

which is the asserted formula.

(iii) The map  $m^{\text{alg}}$  is  $K'$ -multilinear. By additivity in each slot, it suffices to check  $K'$ -homogeneity when every argument is an elementary tensor; there it follows from (ii). The same formula determines the map on all tuples of algebraic tensors, so the initial basis choice does not affect it.

- (iv) For arbitrary decompositions  $u_i = \sum_j x_{ij} \otimes \mu_{ij}$ , multilinearity and (ii) give

$$m^{\text{alg}}(u) = \sum_{j_1, \dots, j_k} m(x_{1j_1}, \dots, x_{kj_k}) \otimes \prod_i \mu_{ij_i}.$$

The cost of this tensor decomposition is at most  $\|m\| \prod_i (\sum_j \|x_{ij}\| \|\mu_{ij}\|)$ . Taking the infimum separately over the decompositions in each slot proves

$$\|m^{\text{alg}}(u_1, \dots, u_k)\|_\pi \leq \|m\| \prod_i \|u_i\|_\pi.$$

(v) To prove strong alternation, suppose two distinct slots  $a, b$  both contain  $u = \sum_j x_j \otimes \mu_j$ . Use this *same* decomposition in both slots and expand all arguments into elementary tensors. Terms using the same index  $j$  in slots  $a$  and  $b$  vanish because  $m$  has a repeated argument. For

$j \neq j'$ , the term using indices  $(j, j')$  in these slots pairs with the term using  $(j', j)$ . Their scalar factors agree, while the values of  $m$  are negatives of one another, so the pair sums to zero. Thus  $m^{\text{alg}}$  vanishes whenever any two arguments coincide. Pair cancellation remains valid in characteristic 2; merely checking  $m^{\text{alg}}(u, \dots, u) = 0$  would not establish strong alternation for  $k \geq 3$ .

(vi) Regard  $m^{\text{alg}}$  as taking values in the completion  $F$ . By (iii), (iv) and Lemma 41(2), it extends to a continuous  $K'$ -multilinear map  $m_{K'}: E^k \rightarrow F$  with  $\|m_{K'}\| \leq \|m\|$ . Strong alternation also passes to the limit: if  $v_a = v_b$ , approximate this common vector by one sequence in  $(E_1)_{K'}$ , and approximate the other entries separately. Every approximating tuple has equal entries in slots  $a, b$ , so (v) and continuity give  $m_{K'}(v) = 0$ .

(vii) Taking every scalar factor equal to one in (ii) gives

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = m^{\text{alg}}(x_1 \otimes 1, \dots, x_k \otimes 1) = m(x) \otimes 1 = \iota_F(m(x)).$$

The defining formula is  $K_1$ -linear in  $m$ . These linear identities pass to the continuous extensions by uniqueness, proving  $K_1$ -linearity of  $m \mapsto m_{K'}$ .  $\square$

**Proposition 45** (descent of a lift). *Assume (H1) and (H2). Let  $E_1$  be a normed  $K_1$ -space,  $F_1$  a  $K_1$ -Banach space, and  $E, F$  as above. If  $Q_{E,E,F}^{K'}$  admits a bounded  $k$ -linear lift  $P$  over  $K'$ , then  $Q_{E_1,E_1,F_1}^{K_1}$  admits a bounded  $k$ -linear lift  $P_1$  over  $K_1$  with  $\|P_1\| \leq \|P\|$ . Equivalently, by Proposition 8 over  $K_1$  and over  $K'$ : if  $Q_{E_1,E_1,F_1}^{K_1}$  is analytic at no point, then  $Q_{E,E,F}^{K'}$  is analytic at no point either.*

*Proof.* Extend the operator and form inputs to  $K'$ , apply the given lift, and retract the value to  $F_1$ . Explicitly, for  $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$ ,  $m \in \text{Alt}_{K_1}^k(E_1; F_1)$  and  $x \in E_1^k$ , set

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F \left( P((f_1)_{K'}, \dots, (f_k)_{K'})(m_{K'})(\iota_E x_1, \dots, \iota_E x_k) \right).$$

We check the norm bound and then the diagonal identity.

For fixed  $f$  and  $m$ , the map  $P((f_i)_{K'})(m_{K'})$  is a continuous alternating  $K'$ -multilinear form on  $E$ . Precomposing each slot with the  $K_1$ -linear isometry  $\iota_E$  and postcomposing with  $\Pi_F$  gives a  $K_1$ -multilinear map on  $E_1$ , still vanishing whenever two arguments coincide. Lemmas 42–44 give

$$\|P_1(f)(m)\| \leq \|\Pi_F\| \|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \leq \|P\| \prod_i \|f_i\| \|m\|.$$

In particular,  $P_1(f)(m) \in \text{Alt}_{K_1}^k(E_1; F_1)$ . The form extension is  $K_1$ -linear in  $m$ , the operator extension is  $K_1$ -linear in each  $f_i$ , and  $P$  is  $K'$ -multilinear with values in  $K'$ -linear operators. Consequently  $P_1$  is  $K_1$ -linear in  $m$  and  $K_1$ -multilinear in  $(f_1, \dots, f_k)$ . The displayed estimate proves boundedness at both levels and yields  $\|P_1\| \leq \|P\|$ .

On the diagonal, use  $P(g, \dots, g) = Q^{K'}(g)$  with  $g = f_{K'}$ , followed by the embedding identities in Lemmas 43 and 44 and the retraction identity  $\Pi_F \iota_F = \text{id}$ :

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F(m_{K'}(f_{K'} \iota_E x_1, \dots, f_{K'} \iota_E x_k)) \\ &= \Pi_F(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)) \\ &= \Pi_F(\iota_F(m(f x_1, \dots, f x_k))) \\ &= m(f x_1, \dots, f x_k) = Q^{K_1}(f)(m)(x). \end{aligned} \quad \square$$

*Remark 46.* Proposition 45 is characteristic-free and uses no hypothesis on  $k$ . It transports negative results upward only, from  $K_1$  to  $K'$ . Ingleton's theorem is used only on  $K'$ , so  $F_1$  may be, and in our application is, non-ultrametric. The completeness of  $F_1$  is used essentially: it is what makes the retraction  $\Pi_F$  land in  $F_1$ .

**7.4. Incomplete base fields.** The preceding construction takes place over the completion  $\widehat{K}$  of the prescribed field. To return to  $K$ , we restrict scalars on the resulting Banach spaces. Density of  $K$  in  $\widehat{K}$  forces every continuous  $K$ -multilinear map between these spaces to be  $\widehat{K}$ -multilinear, so a lift cannot appear under this restriction.

**Lemma 47.** *Let  $K$  be a nontrivially normed field with completion  $\widehat{K}$ , and let  $E, E', F$  be  $\widehat{K}$ -Banach spaces, regarded as  $K$ -Banach spaces by restriction of scalars (with the same norms). Then  $\mathcal{L}_K(E, E') = \mathcal{L}_{\widehat{K}}(E, E')$ ,  $\text{Alt}_K^k(E; F) = \text{Alt}_{\widehat{K}}^k(E; F)$  and  $\text{Alt}_K^k(E'; F) = \text{Alt}_{\widehat{K}}^k(E'; F)$  with the same norms,  $Q_{E, E', F}^K = Q_{E, E', F}^{\widehat{K}}$ , and every bounded  $k$ -linear lift of  $Q_{E, E', F}^K$  over  $K$  is a bounded  $k$ -linear lift of  $Q_{E, E', F}^{\widehat{K}}$  over  $\widehat{K}$ .*

*Proof.* Let  $g$  be a continuous  $K$ -linear map between normed spaces that already carry compatible  $\widehat{K}$ -scalar actions. For  $\lambda \in \widehat{K}$ , choose  $\lambda_n \in K$  with  $\lambda_n \rightarrow \lambda$ . Since  $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda| \|x\|$ , continuity gives

$$g(\lambda x) = \lim_n g(\lambda_n x) = \lim_n \lambda_n g(x) = \lambda g(x).$$

Thus  $g$  is  $\widehat{K}$ -linear. Applying the same argument with all but one slot fixed shows that every continuous  $K$ -multilinear map is  $\widehat{K}$ -multilinear. Alternation is the same equal-input vanishing condition over either field.

The spaces of bounded maps therefore coincide as sets. Their operator norms agree as well, since the norms are defined by the same pointwise bounds on the same vectors. They are  $\widehat{K}$ -Banach spaces, and  $Q_{E, E', F}^K$  and  $Q_{E, E', F}^{\widehat{K}}$  agree by their defining precomposition formula. Finally, apply the slotwise density argument once more to a bounded  $k$ -linear lift  $P$  between these map spaces. It becomes  $\widehat{K}$ -multilinear with the same norm and the same diagonal identity, and hence is a lift over  $\widehat{K}$ .  $\square$

## 8. PROOFS OF THE MAIN RESULTS

The preceding sections give the obstruction to a bounded lift. We first establish the remaining property of the target, then assemble the proof of Theorem 1 and deduce Corollary 2.

### 8.1. The target has no equivalent ultrametric norm.

**Proposition 48.** *Let  $\kappa$  be any field,  $k \geq 2$ ,  $r \in (0, 1)$ , and let  $K_1 = \kappa((X))$ ,  $E_1$ ,  $E_0$  and  $B$  be as in §6. Let  $e_i \in E_0$  be the unit vectors and  $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{kj+k} \in B$  ( $j \geq 0$ ), wedges on pairwise disjoint blocks of indices. Then  $\|\omega_j\|_B = 1$  and*

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq N \quad (N \geq 1).$$

*Hence the norm of  $B$  is not ultrametric, and  $B$  admits no equivalent ultrametric norm. The same holds for  $F = B \widehat{\otimes}_{\pi} K'$  as in §7, for any  $K' \supseteq K_1$  satisfying (H1) and (H2), regarded as a normed space over  $K'$  or over any subfield of  $K'$ .*

*Proof.* Each  $\omega_j$  is a wedge of unit vectors, so  $\|\omega_j\|_B \leq 1$  by the defining projective norm. Evaluation of its determinant array at the block indices gives  $\det I = 1$ , hence  $\|\omega_j\|_B \geq \|J\omega_j\|_{\infty} \geq 1$ . The triangle inequality now gives the upper bound  $N$ .

For the lower bound, write  $\xi_N$  for the sum of the same block wedges in  $\Lambda_{\kappa}^k E_0$ , and put  $\omega := \sum_{j < N} \omega_j \in B$ . Its array  $J\omega$  is the constant Laurent array  $\Omega^{\kappa}(\xi_N)$ . Thus  $\text{coeff}_0(J\omega) = \Omega^{\kappa}(\xi_N)$ , and uniqueness in Proposition 31 gives  $\eta(\omega) = \xi_N$ .

We apply Lemma 13 to  $\xi_N$ . Fix one of the first  $N$  blocks and an index  $i_0$  in it. Take for  $\varphi$  the  $k-1$  coordinate functionals corresponding to the other indices of that block. On every other block all these functionals vanish, so its contraction is zero; here  $k \geq 2$  ensures that the minors have positive size. On the chosen block, the coefficient of  $e_{i_0}$  is  $\pm 1$ , while every other cofactor contains the zero column obtained by evaluating the functionals at  $e_{i_0}$ . Consequently  $c_\varphi(\xi_N) = \pm e_{i_0}$ . These contractions span all  $kN$  unit vectors in the first  $N$  blocks, so

$$kN \leq \text{sdim}(\xi_N) = \text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_B$$

by (6.2). This proves the lower bound.

If  $\|\cdot\|_u$  were an equivalent ultrametric norm, choose positive constants  $c, c'$  with  $\|\cdot\|_B \leq c\|\cdot\|_u \leq c'\|\cdot\|_B$ . Then

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq c \max_{j < N} \|\omega_j\|_u \leq c'$$

for every  $N \geq 1$ , a contradiction. In particular, the original norm is not ultrametric. This argument uses only the ultrametric inequality for  $\|\cdot\|_u$ , not its scalar homogeneity.

Finally,  $\iota_F: B \rightarrow F$  is a  $K_1$ -linear isometry by Lemma 42(3). The images  $\iota_F \omega_j$  satisfy the same unit-norm and sum-growth estimates, so the argument applies to  $F$  as well. Restriction to a subfield leaves these estimates unchanged.  $\square$

If  $r \leq \frac{1}{2}$ , then  $M_r = 1$  and  $\|\sum_{j < N} \omega_j\|_B = N$  exactly.

## 8.2. Proof of Theorem 1.

*Proof.* Let  $K$  be a nontrivially normed field of characteristic  $p > 0$ , and let  $k \geq p$ .

- (1) *Choose a Laurent subfield.* By Lemma 6,  $K$  is ultrametric. Its completion  $\widehat{K}$  is a complete, nontrivially normed, ultrametric field of characteristic  $p$ . Choose  $t \in K$  with  $0 < |t| = r < 1$ , and let  $K_1$  be the closure of  $\mathbb{F}_p(t)$  in  $\widehat{K}$ . Lemma 37 identifies  $K_1$  isometrically with  $\mathbb{F}_p((X))$ , with  $|X| = r$ , by  $\text{ev}_t$ . In particular,  $K_1$  is complete and spherically complete.
- (2) *Obtain the obstruction over  $K_1$ .* Put  $E_1 := \ell^\infty(\mathbb{N}, K_1)$ , and let  $B$  be the completion of  $\Lambda_{K_1}^k E_1$  under the projective exterior norm. Transport of scalars along  $\text{ev}_t$  preserves the norms, exterior powers and completions in this construction. It also identifies the operator and alternating-map spaces, their precomposition maps, and bounded lifts. Theorem 33, with  $\kappa = \mathbb{F}_p$ , therefore shows that  $Q_{E_1, E_1, B}^{K_1}$  has no bounded  $k$ -linear lift.
- (3) *Ascend to  $\widehat{K}$ .* The inclusion  $K_1 \subseteq \widehat{K}$  satisfies (H1) and (H2), and  $B$  is complete. Proposition 45, with  $K' = \widehat{K}$ , gives Banach spaces

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \quad F := B \widehat{\otimes}_\pi \widehat{K}$$

for which  $Q_{E, E, F}^{\widehat{K}}$  has no bounded  $k$ -linear lift.

- (4) *Descend to  $K$ .* Regard the same  $E$  and  $F$  as  $K$ -Banach spaces. By Lemma 47, a bounded lift of  $Q_{E, E, F}^{\widehat{K}}$  would extend to a bounded lift of  $Q_{E, E, F}^{\widehat{K}}$ , which is impossible. Proposition 8 applies over the possibly incomplete field  $K$ :  $Q_{E, E, F}^{\widehat{K}}$  is analytic at no point, and  $\text{ContDiffAt } K \ \omega$  fails at every point. Reindexing as in §3 gives the same conclusions, with the same spaces  $E, F$ , for every finite  $\iota$  with  $|\iota| = k$ .
- (5) *Exclude an equivalent ultrametric target norm.* Since  $k \geq p \geq 2$ , Proposition 48 applies to  $B$  and  $F$ , after the scalar identification in step (2). Its conclusion persists when  $F$  is regarded over  $K$ .

$\square$

*Remark 49.* If  $K_1 = \widehat{K}$ , as when  $K = \mathbb{F}_p((u))$  and  $t = u$ , the completed tensor products identify isometrically with  $E_1$  and  $B$ : use Lemma 42(3) and  $x \otimes \lambda = \lambda x \otimes 1$ . Thus  $E$  is ultrametric in this case. For  $K = \mathbb{F}_q((u))$ ,  $q = p^n$ , with its  $u$ -adic absolute value, one can instead apply Theorem 33 directly with  $\kappa = \mathbb{F}_q$  and  $X = u$ , obtaining  $E = \ell^\infty(\mathbb{N}, K)$  and  $F = B$ . The proof over a general base does not require, or assert, that  $E$  is ultrametric.

### 8.3. Proof of Corollary 2.

*Proof.* Put  $k := |\iota|$ . If  $k! \neq 0$  in  $K$ , Proposition 14 gives a finite power series at every point for arbitrary normed  $E, E', F$ , including degree zero. If  $k! = 0$  in  $K$ , the field has positive prime characteristic  $p \leq k$ . Theorem 1 gives the required Banach counterexample. Together these prove the classification and both strengthened conclusions.  $\square$

## 9. COMPLEMENTS AND OPEN PROBLEMS

We apply Theorem A to finite-dimensional targets and to targets over discretely valued fields, then give an explicit lift from an orthogonal basis. The section ends with comparisons and questions about ultrametric targets.

### Finite-dimensional targets.

*Remark 50* (finite-dimensional targets). Let  $K$  be a complete nontrivially normed field, let  $E, E'$  be normed  $K$ -spaces, and let  $F$  be finite-dimensional. If  $F \neq 0$ , then, at every  $f_0 \in \mathcal{L}(E, E')$ ,

$$Q_{E, E', F}^K \text{ is analytic at } f_0 \iff Q_{E, E', K}^K \text{ is analytic at } f_0.$$

If  $F = 0$ , precomposition is constant and hence analytic.

Indeed, choose a basis  $v_1, \dots, v_d$  of  $F$  with coordinate functionals  $\lambda_1, \dots, \lambda_d$ . These functionals are continuous because  $K$  is complete. For a vector  $v$  and a scalar-valued form  $a$ , write  $va$  for the form  $x \mapsto a(x)v$ . Writing  $Q_F := Q_{E, E', F}^K$  and  $Q_K := Q_{E, E', K}^K$ , we have

$$Q_F(f)(m) = \sum_{a=1}^d v_a Q_K(f)(\lambda_a \circ m).$$

The right-hand side is a fixed continuous linear function of the operator  $Q_K(f)$ , so analyticity of  $Q_K$  implies that of  $Q_F$ . Conversely, choose  $v \in F$  and  $\lambda \in \mathcal{L}(F, K)$  with  $\lambda(v) = 1$ . For  $a \in \text{Alt}^k(E'; K)$ ,

$$Q_K(f)(a) = \lambda \circ Q_F(f)(va).$$

This expresses  $Q_K$  as a fixed continuous linear function of  $Q_F$  and proves the converse. Thus finite-dimensionality of the target reduces the question to a scalar target. This reduction gives no assertion of analyticity over a nonspherically complete base.

If  $K$  is ultrametric and spherically complete, there is a positive conclusion for every finite-dimensional  $F$ , with any norm. The coordinate norm  $\|y\|_\infty := \max_a |\lambda_a(y)|$  is equivalent to the given norm. With this norm,  $F$  is spherically complete: a closed ball in  $K^d$  is a product of coordinate balls, and spherical completeness of  $K$  gives a point in each coordinate intersection. Theorem A therefore applies. Returning to the original norm preserves analyticity, since it changes the norms on the alternating-map and operator spaces only by fixed comparison factors.

### Targets over discretely valued bases.

**Proposition 51** (targets over discretely valued bases). *Let  $K$  be an ultrametric normed field with  $|K^\times| = r^\mathbb{Z}$  for some  $r \in (0, 1)$ , complete or not; for instance  $K = \mathbb{F}_q((t))$  or  $K = \kappa((t))$ . Let  $E, E'$  be normed  $K$ -spaces and  $F$  a  $K$ -Banach space that admits an equivalent ultrametric norm. Then  $Q_{E, E', F}^K$  is analytic at every point.*

*Proof. Round the norm.* Let  $\|\cdot\|_u$  be an equivalent ultrametric norm on  $F$ . For  $y \neq 0$ , round  $\|y\|_u$  up to the least element  $\|y\|_u^\sharp$  of  $r^\mathbb{Z}$  above it, and put  $\|0\|_u^\sharp := 0$ . Then

$$\|y\|_u \leq \|y\|_u^\sharp \leq r^{-1}\|y\|_u.$$

Rounding is monotone and commutes with multiplication by elements of  $r^\mathbb{Z}$ . Since  $|\lambda| \in r^\mathbb{Z}$  for  $\lambda \in K^\times$ , it preserves scalar homogeneity and the ultrametric inequality. Thus  $\|\cdot\|_u^\sharp$  is an equivalent ultrametric norm, and  $F' := (F, \|\cdot\|_u^\sharp)$  is complete with distances in  $r^\mathbb{Z} \cup \{0\}$ .

Apply Theorem A. Lemma 35 makes  $F'$  spherically complete. Theorem A therefore gives analyticity of  $Q' := Q_{E, E', F'}^K$  at every point.

Return to the original norm. Equivalent target norms give the same spaces of bounded alternating maps, with equivalent operator norms. Consequently

$$\mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)) \quad \text{and} \quad \mathcal{L}(\text{Alt}^k(E'; F'), \text{Alt}^k(E; F'))$$

are the same vector space with equivalent norms, and  $Q = Q'$  under this identification. A power series for  $Q'$  is therefore a power series for  $Q$ : coefficient norms change by at most a fixed factor, and convergence is unchanged.  $\square$

Thus, over a discretely valued base, every Banach counterexample must have a target admitting no equivalent ultrametric norm. Proposition 48 establishes this property for our target over every base.

*Remark 52* (necessary, but not sufficient). Completeness of the target is essential to this deduction: it is used through Lemma 35 in Proposition 51. Even for complete targets, the absence of an equivalent ultrametric norm is not sufficient for a counterexample. Take the degree-3 space  $B$  of §6.2 over  $K_1 = \mathbb{F}_5((X))$ . It admits no equivalent ultrametric norm by Proposition 48, which requires only  $k \geq 2$ , but  $Q_{E_1, E_1, B}^{K_1}$  is analytic by Corollary 2(1), since  $3! = 1$  in  $\mathbb{F}_5$ .

The factorial hypothesis is also essential to the finite-field obstruction. Applying the construction of §6.4 to this bounded lift gives a map  $\Psi$  with all the properties of Lemma 32, which is valid in every degree. Thus  $\Psi$  satisfies every hypothesis of Theorem 23(2), with  $L = \mathbb{F}_5$ , except  $k! = 0$ ; compare Remark 29.

### An explicit lift from an orthogonal basis.

*Remark 53* (an explicit lift). Let  $K$  be ultrametric, let  $E$  have an orthogonal basis  $(e_i)_{i \in I}$ , and let  $F$  be complete and ultrametric. Here a basis means that every  $x \in E$  has an unconditional expansion  $x = \sum_i e_i^*(x)e_i$ , with  $\|\sum_i c_i e_i\| = \sup_i |c_i| \|e_i\|$ . Fix a total order on  $I$  and put

$$P(f_1, \dots, f_k)(m)(x^1, \dots, x^k) := \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

For each  $x$ , the weighted coordinates  $|e_i^*(x)| \|e_i\|$  tend to zero off finite subsets of  $I$ . Expanding the determinant bounds a summand by a fixed constant times a finite sum of products of these null families, one coordinate from each input. Such products tend to zero off finite subsets of the tuple space. Hence, for every  $\varepsilon > 0$ , only finitely many summands have norm at least  $\varepsilon$ , and the family is summable in the complete ultrametric space  $F$ .

Orthogonality gives  $|e_s^*(x)| \|e_s\| \leq \|x\|$ . The ultrametric determinant estimate therefore bounds every summand by  $\|m\| \prod_a \|f_a\| \prod_b \|x^b\|$ ; the ultrametric inequality in  $F$  then gives  $\|P\| \leq 1$ . The formula is multilinear in  $f$  and linear in  $m$ , and its output is alternating in  $x$  because the determinant is alternating in its columns. Finally, finite coordinate expansions give

$$m(fx^1, \dots, fx^k) = \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(fe_{s_1}, \dots, fe_{s_k});$$

the general identity follows by continuity along finite coordinate projections. Thus  $P(f, \dots, f) = Q(f)$ . For  $k = 0$ , the formula has one empty-index term, with determinant 1.

This *sorted lift* works in every characteristic and degree, without division by  $k!$  or spherical completeness. It is not used in the main proof. Over a complete discretely valued field, every ultrametric Banach space is topologically isomorphic to a space  $c_0(I)$  of null families [9, §1, Prop. 1 and its Corollaire], and hence has such a basis for an equivalent norm.

### Ultrametric counterexamples.

*Remark 54* (ultrametric targets over bases that are not spherically complete; announced). The following statement is announced without proof and is not used elsewhere in this paper. Let  $k \geq 2$ , and let  $K$  be a complete nontrivially normed field, not spherically complete, with  $k! = 0$  in  $K$ . Then  $K$  is ultrametric by Lemma 6. For a cardinal  $\mu$  put  $\beth_0(\mu) := \mu$  and  $\beth_{n+1}(\mu) := 2^{\beth_n(\mu)}$ , and let  $\mu^+$  be the successor cardinal of  $\mu$ . Let  $I$  be a set of cardinality at least  $\beth_{3k-1}(|K|)^+$ , put  $E := \ell^\infty(I, K)$ , and let  $F_I$  be the closure in  $\ell^\infty(I^k, K)$  of the span of the determinant arrays  $(i_1, \dots, i_k) \mapsto \det(x_b(i_a))_{a,b}$ ,  $x \in E^k$ . Then  $E$  and  $F_I$  are ultrametric  $K$ -Banach spaces, and  $Q_{E,E,F_I}^K$  is analytic at no point. The proposed proof uses a partition argument with the Erdős–Rado theorem in place of Ramsey’s theorem. For  $k = 2$  in characteristic 2 the countable index set  $\mathbb{N}$  suffices [6].

Comparison with characteristic 2. For  $p = k = 2$ , Theorem 1 gives a second counterexample over every nontrivially normed field of characteristic 2. Unlike the construction of [6], it applies to spherically complete bases such as  $\mathbb{F}_2((t))$ . The distinction lies in the target: that of [6] is ultrametric, whereas ours admits no equivalent ultrametric norm. Proposition 51, using Theorem A, shows that a complete counterexample target over  $\mathbb{F}_2((t))$  cannot carry an equivalent ultrametric norm.

The proofs also use different mechanisms. The characteristic-2 argument combines a parity identity for determinants—the determinant of an alternating matrix is the square of its Pfaffian—with the failure of Hahn–Banach over a base that is not spherically complete, using  $\ell^\infty$  to assemble the construction. Here the finite-field theorem is combined with the constant-coefficient map of a Laurent series field, and Ingleton’s theorem is applied over a spherically complete subfield. In characteristic 2, Theorem 23 needs no signs (Remark 28).

### An alternative algebraic approach.

*Remark 55* (another possible approach). The following optional sketch suggests an alternative route to the input needed in §6. It requires two additional results: an infinite-field obstruction and an equality between support dimension and contraction rank. Neither is proved here, and no argument in this paper depends on this route.

*The infinite-field input.* The first required statement is the analogue of Theorem 23 over infinite fields  $L$  with  $k! = 0$  in  $L$ , for lifts that satisfy (Pw) and have uniformly bounded support, of an arbitrary linear family  $D: A \rightarrow \text{End}_L(V)$  whose image modulo the finite-rank operators is infinite-dimensional.

*Base change and minors.* Let  $\Psi$  be a lift over  $\mathbb{F}_q$  whose values have support dimension at most  $d$ . After the finite base change  $\mathbb{F}_q \subseteq \mathbb{F}_{q^N}$ , the values of the base change of  $\Psi$  at  $\mathbb{F}_{q^N}$ -points have support dimension at most  $d_N := N^{2k}d$ , since each input has only  $N$  pieces over  $\mathbb{F}_q$ ; choose  $N$  with  $q^N > d_N + 1$ . One would then use that the support dimension of  $\omega \in \Lambda^k V$  equals the rank of its contraction matrix, the matrix of coordinates of the vectors  $c_\varphi(\omega)$  of Lemma 13, with  $\varphi$  running over tuples of dual basis vectors. Lemma 13 gives only the inequality  $\geq$ , and *we do not prove the equality*. Granting it, consider any finite set of independent scalar parameters in each of the  $2k$  input slots. By multilinearity, every contraction-matrix entry has degree at most one in each scalar parameter. Its  $(d_N + 1)$ -minors therefore have degree at most  $d_N + 1 < q^N$  in each variable. They vanish at every  $\mathbb{F}_{q^N}$ -point, so they are zero polynomials and remain zero over the algebraic closure  $\overline{\mathbb{F}_q}$ . The resulting support bound would allow the proposed obstruction to be applied over this infinite field.

*Compatibility requirements.* In addition to those two unproved inputs, the route requires the following properties of  $\Psi$ .

- (1) The full identity (Pol), not only (Pol1): for  $k > q$  the identity (Pw) at  $\mathbb{F}_q$ -points does not survive base change (Proposition 22(3)), whereas (Pol) gives (Pw) at every point after base change.
- (2) Alternation, as required by the unproved input above. Both alternation and antisymmetry survive scalar extension, but in characteristic 2 antisymmetry does not imply alternation.
- (3) An infinite-dimensional image modulo finite rank. For the multipliers  $D_a x = ax$  on  $L^\mathbb{N}$  this holds because the multipliers by the indicators of pairwise disjoint infinite sets are linearly independent modulo finite rank.

The lift of §6 satisfies (Pol) and is alternating (Lemma 32), and its multiplier family is the one in (3), with  $L = \kappa$ .

**Open problems.** The questions left open concern ultrametric targets.

- (1) *Countable index.* Let  $K$  be complete, not spherically complete, of characteristic  $p \geq 3$ , and  $k \geq p$ . Is there a complete ultrametric  $F$  such that  $Q_{E,E',F}^K$  is not analytic for  $E = \ell^\infty(\mathbb{N}, K)$ ? For  $p = 2$  the answer is yes, by [6]: for  $k = 2$  this is the theorem of [6], and for  $k \geq 3$  it follows by the wedging argument of [6, Remark 18], which is formalized in the Lean theorem of §10 under the hypothesis (K2) of [6]. Remark 54 announces a counterexample with a much larger index set, leaving the countable case open.
- (2) *Densely valued spherically complete bases.* Let  $K$  be spherically complete, densely valued and of characteristic  $p$ , for example the Hahn series field  $\mathbb{F}_p((t^\mathbb{Q}))$ , and let  $k \geq p$ . Is  $Q_{E,E',F}^K$  analytic for all Banach  $E, E'$  and every complete ultrametric  $F$ ? Theorem A gives a positive answer when  $F$  is spherically complete. The counterexample of Theorem 1 has a non-ultrametric target and does not answer this question.

## 10. FORMALIZATION

The proof snapshot of this paper is recorded in the private repository [11], at commit [5e9c3cf](#), using Lean 4.34.0-rc2 and Mathlib revision [2b73d9821d29](#). At that revision, the proof library was built without admitted proofs, and its transitive axiom check reported only `propext`, `Classical.choice` and `Quot.sound`. The repository's `THEOREM_MAP.md` maps the paper's statements to the Lean declarations.

The principal declarations, in the namespace `AlternatingAnalytic`, are

- `exists_banach_counterexample_full`, for Theorem 1, including incomplete base fields, the same Banach witnesses for every finite index type of cardinality  $k$ , no bounded lift, nowhere analyticity, nowhere  $C^\omega$ , and no equivalent ultrametric norm on the target;
- `factorial_ne_zero_iff_allBanachPrecompositionAnalytic`, for both directions of Corollary 2;
- `finiteField_multiplier_theorem`, for both parts of Theorem 23, with the support bound required only on the stated finitely supported test inputs in its first part.

The supporting development covers the  $\kappa$ -linear coefficient map and support estimate (Proposition 31), all four properties of the residue lift (Lemma 32), the Laurent counterexample, and ascent and descent to a prescribed field. It also includes the lift criterion, the sharp polarization distinctions, Theorem A, Proposition 51, and the sorted lift of Remark 53. Spherical completeness uses the development’s own class, with the definition in §2. The formalization used Codex with Autoform and incorporated previously formalized ingredients with their provenance retained.

Proposition 19 reproduces the argument of [7], whose formal proof is included in the pinned Mathlib dependency. The finite-coordinate proof (Proposition 15 and Corollary 16) and the finite-target reduction (Remark 50) are supplementary informal results; they are not claimed as separate formalized declarations in this development. The categorical formulations in the introduction are mathematical corollaries of the formalized operator results, rather than separate Lean categorical declarations.

The characteristic-2 counterexample [6] has separate provenance. Its formalization proves non-analyticity in degree 2 at the origin, with a wedging extension under (K2) to every degree at least 2 and every base point: `Round24Transfer.char2_not_analyticAt_of_hasEmptyNestedBalls`. That construction is not re-proved or re-audited here.

The proposed alternative route of Remark 55, the announced large-index construction of Remark 54, the open problems, and Serre’s external classification cited in Remark 53 are excluded from the proved results. The repository’s comparator challenge contains two short contradiction corollaries of the main theorem and Theorem A for every regularity order. Their solutions use the corresponding library theorems. The Banach and spherical-completeness hypotheses are given by transparent definitions in Mathlib terms. Intentional placeholders occur only in the challenge file, not in the proof library or solution. Current challenge checks use Kim Morrison’s comparator, with the configuration recorded in the repository.

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